

Review

A comparison review of transfer learning and self-supervised learning: Definitions, applications, advantages and limitations

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ABSTRACT

Deep learning has emerged as a powerful tool in various domains, revolutionising machine learning research. However, one persistent challenge is the scarcity of labelled training data, which hampers the performance and generalisation of deep learning models. To address this limitation, researchers have developed innovative methods to overcome data scarcity and enhance deep model learning capabilities. Two prevalent techniques that have gained significant attention are transfer learning and self-supervised learning. Transfer learning leverages knowledge learned from pre-training on a large-scale dataset, such as ImageNet, and applies it to a target task with limited labelled data. This approach allows models to benefit from the learned representations and effectively transfer knowledge to new tasks, resulting in improved learning performance and generalisation. On the other hand, self-supervised learning focuses on training models using pretext tasks that do not require manual annotation, allowing them to learn valuable representations from large amounts of unlabelled data. These learned representations can then be fine-tuned for downstream tasks, mitigating the need for extensive labelled data. In recent years, transfer and self-supervised learning have found applications in various fields, including medical image processing, video recognition, and natural language processing. These approaches have demonstrated remarkable achievements, enabling breakthroughs in areas such as disease diagnosis, object recognition, and language understanding. However, while these methods offer numerous advantages, they also have limitations. For example, transfer learning may face domain mismatch issues between the pre-training and target domains, while self-supervised learning requires careful design of pretext tasks to ensure meaningful representations. This review paper explores the recent applications of these pre-training methods in various fields within the past three years. It delves into the advantages and limitations of each approach, assesses the performance of models employing these techniques, and identifies potential directions for future research. By providing a comprehensive review of current pre-training methods, this article offers guidance for selecting the best technique for specific deep learning applications to address the data scarcity issue.

1. Introduction

During the past few years, machine learning (ML) has improved significantly with the appearance of Deep Learning technology (LeCun, Bengio, & Hinton, 2015). Usually, deep learning models are built in a supervised learning framework to enable their efficiency in learning good data representations, in which a model is trained from scratch to optimise the performance of a single task and aims to gain the ability to generalise unseen test data. The remarkable advancement in hardware systems has led to the emergence of new studies in deep learning. It makes it quickly be considered the first choice when building intelligent systems in classification (Wang, Fan, & Wang, 2021), detection (Ma

et al., 2023), segmentation (Liu et al., 2023; Minaee et al., 2021), and other computer vision tasks (Alzubaidi et al., 2021). So far, most of the achievements that deep learning approaches have achieved are based on large scales of training data, which always require millions of training data to help the model achieve satisfactory performance. Bringing factors such as time consumption, expense cost, and data quality, a well-annotated dataset is difficult to access in many real-world cases (Bansal, Sharma, & Kathuria, 2022), and lack of sufficient training data may lead to poor prediction performance and generalised capability for the new scenarios, and lead to rapid overfitting of the training model. To address this significant challenge and promote the

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development of deep learning technology, more and more efforts have been devoted to developing various approaches (Alzubaidi et al., 2023).

Among all the different methods that aim to alleviate deep learning's data scarcity problem, the pre-training method has been considered a central technique when conducting deep learning research and applications; the critical method for pre-training is to learn effective representations from outer significant source datasets and transfer the extracted knowledge from the source domain to the target domain to help accelerate target task learning. As a method to make use of existing knowledge instead of training from scratch, it is well known that pre-training reduces the deep learning model's dependency on the size of the training dataset and improves the model's generalised ability when the target domain's dataset is minimal (Hendrycks, Lee, & Mazeika, 2019). Transfer learning (TL), also known as knowledge transfer, is one of the most popular pre-training methods. The training process of Transfer Learning (TL) involves initially training a baseline model on a large dataset and subsequently fine-tuning it using a smaller target dataset to achieve optimal performance. Instead of directly training the model on the relatively limited target dataset, TL leverages parameters pre-trained on other source data to establish a robust foundation model. Empirical evidence from previous experiments demonstrates that TL approaches effectively combat overfitting, expedite training, and notably reduce the dataset size required for retraining. Furthermore, TL methods have played a pivotal role in advancing feature fusion research, delivering more precise and diverse feature sets (Al-Timemy et al., 2023; Alammari et al., 2023; Jebur, Hussein, Hoomod, & Alzubaidi, 2023). Currently, TL is widely regarded as the preferred choice in numerous domains for harnessing deep learning technology. Similar to TL, another new pre-training method, self-supervised learning (SSL) has emerged to extract reusable features from source data and utilise them to enhance the training of new models. A significant distinction of SSL lies in its foundation in unsupervised learning techniques. SSL's pre-training phase heavily relies on data augmentation methods to generate synthetic data, which serves as pseudo labels and provides meaningful guidance during training. The subsequent retraining phase of SSL is grounded in supervised learning approaches and typically does not necessitate additional fine-tuning steps. Remarkably, SSL methods can achieve performance on par with many strongly supervised models without the need for additional manually annotated labels, which has recently propelled its popularity. To date, SSL-related research has showcased promising performance in solving data scarcity problems across a wide spectrum of application fields (Yu et al., 2023), and even excelled when confronted with exceptionally small datasets, which makes it a compelling choice when creating deep learning models.

Despite some optimistic research predicting that self-supervision pre-training has the possibility to eventually augment or replace the use of fully supervised pre-training (Doersch & Zisserman, 2017), from recent specific applications, the model using the TL method still outperforms those using the SSL method and the other way around in other applications. As Yang et al. (2020) mentioned in their research, there is a lack of studies on the examination of which properties of data and tasks make one approach successful and surpass the other. As a result, machine learning researchers waste unnecessary resources and time in performance comparison when selecting the suitable pre-training method.

As mentioned above, and to the best of our knowledge, a minimal number of papers exist to discuss the discrepancies between TL and SSL technologies. Researchers have not yet adequately answered how to choose the appropriate pre-training method. Consequently, we have set three research questions in this review to better understand the current advantages and limitations of TL and SSL and guide future research. The research questions are shown below:

1. How do TL and SSL work, and what is the difference between these two methods?
2. What are the advantages and limitations of the TL and SSL in different applications?

3. What are the future directions for solving or avoiding the limitations of TL and SSL?

To address these research questions, we have conducted this literature review to learn from recent state-of-the-art TL/SSL applications to compare and summarise their advantages and limitations in different application areas. The methodology for this review has been proposed in the following section, including a description of the source and keywords and thoroughly developed inclusion/exclusion criteria based on our research questions. We have reviewed 128 articles in this survey, including 36 articles on TL applications, 44 articles on SSL applications, and 48 articles on relevant seminal research and review published within the last four years.

The following outlines the main contributions of our study.

1. This research summarised the main definitions for the TL and SSL approaches and listed their differences.
2. This survey reviewed recent TL and SSL applications to shed light on state-of-the-art architectures. The performance of TL/SSL approaches in different fields has been compared to provide suitable recommendation methods in different application areas.
3. This study discussed the current research gaps between these two methods and summarised the future directions of recent research.

The rest of this paper is structured as follows: Section 2 presents the review methodology employed in this study. Section 3 provides a comprehensive analysis of the distribution of papers across different years, sources, and the correlations between keywords. Section 4 offers definitions for the TL and SSL technologies and an overview of their main categories. Section 5 compares the two methods, highlighting the distinctions in their training processes. Section 6 focuses on recent applications of TL and SSL approaches, offering an analysis of their performance. Section 7 summarises the advantages and limitations of the TL and SSL methods. Section 8 presents a performance comparison between the two methods, offers recommendations for selecting suitable pre-training methods for different application fields, and discusses the reasons and necessity of improving the two methods in future work. The final section (Section 9) summarises the findings of our study and discusses future directions for the TL and SSL methods.

2. Review methodology

This section introduces the methodology used to perform the literature review, including determining the literature sources, search keywords, inclusion/exclusion criteria, and paper selection criteria. In this review, we have reviewed several research papers on TL and SSL technologies, aiming to summarise their current applications, advantages and limitations, and future research directions.

2.1. Sources and keywords

We focused our literature search on journals and conferences from these most reputed publishers to allow credibility, and we consider four bibliographic databases as primary sources, namely:

- IEEE Explore (<http://ieeexplore.ieee.org/>)
- ACM Digital Library (<https://dl.acm.org/>)
- Springer (<https://link.springer.com/>)
- Scopus (<https://www.scopus.com/>)
- Google Scholar (<https://scholar.google.com/>)

We selected these databases as our sources not only for their high reputation but also because the articles cited within them have undergone rigorous peer review, ensuring that the work is grounded in original research. Each of these databases offers unique strengths: IEEE Xplore provides comprehensive access to the technical and scientific literature, offering full-text papers in the domains of electrical engineering, electronics, and computer science. ACM specialises in housing academic professional papers within the realm of computer science, making it a prime resource for our research. Springer boasts

an extensive repository of papers spanning diverse fields, including biomedicine, psychology, education, policy, and science. Scopus hosts a wealth of resources in fields such as medicine, health, science, and engineering, further enhancing the comprehensiveness of our research. Collectively, these databases furnish a solid foundation for our literature review, encompassing a wide spectrum of research domains and ensuring the credibility and relevance of the papers we have incorporated.

Additionally, we also considered remarkable preprint works as our support criteria. The ArXiv database was used as our secondary source:

- ArXiv Preprints (<https://arxiv.org/>)
- The related works sections in the selected papers

It is important to note that preprint publishers like ArXiv often feature work that has not undergone full peer review, which can affect their reliability. Therefore, in our review, we exercise caution when considering such work. We prioritise preprints that have garnered substantial citations from published papers, as this citation history can serve as an indicator of their credibility and information accuracy.

Additionally, the keywords used for search criteria are listed as follows:

("Deep Learning"), ("Transfer learning"), ("Self-supervised learning"), (("Deep learning") AND ("Data scarcity")), ("Deep learning" AND ("Transfer learning" OR "Self-supervised learning")), (("Transfer learning" OR "Self-supervised learning") AND ("Application" OR "SOTA")).

The selection of keywords, namely 'deep learning' and 'data scarcity,' serves as a deliberate choice to pinpoint studies that specifically address data limitations. This is in line with the primary motivation to investigate pre-trained learning methods. Additionally, keywords such as 'transfer learning', 'self-supervised learning', and 'application' have been chosen to narrow our focus to TL and SSL approaches, excluding other pre-trained methods. This strategic selection of keywords helps to emphasise the core theme of our study.

2.2. Inclusion and exclusion criteria

Our inclusion criteria were designed to ensure the reliability and relevance of selected articles. We initially focused on research work written in English and published in reputable journals or conference proceedings. These papers were expected to address the scarcity of deep learning data or demonstrate the real-world applications of TL and SSL. Specifically, we explore the literature on TL and SSL from 2020 to 2023, as this is when SSL starts to attract attention and becomes widely applied in different areas. This is because our goal is to analyse the recent developments in TL and SSL and to emphasise their advantages and limitations. Furthermore, the titles and abstracts of the research articles were examined to judge the relevance of the search results. We considered research works focused on applying TL or SSL methods to solving real-world problems and have proposed novel approaches in their fields. We also include those significant research and survey papers that have been frequently cited, even if they are not within the predefined period, as they could represent important milestones in technique development. In addition, we added some methods that first introduced the TL and SSL methods in some areas to explore their efficiency and generalisability in different contexts.

Our exclusion criteria aimed to refine our selection and ensure the quality of chosen papers. We kept refining our search results by selecting papers that have received relatively high citation numbers in different databases, for example, more than ten times of citations in one database to make sure the selected papers have been examined and are worth trust. Besides, we removed all duplicated papers that occurred more than once from various literature sources. Additionally, we have excluded any other work outside our threshold in publishing years to maintain the review's up-to-date nature. Later, we filtered out several research works that are less related to our search topics, for example, using TL or SSL approaches just for comparison with

other deep learning algorithms or focusing on theoretical improvements without considering real-world problem-solving. For ArXiv preprints, we only considered works that have received high citations and have been cited by published papers.

Apart from all those inclusion and exclusion criteria we mentioned above, we also included a few works from outside the selected databases that may not fit the inclusion standard but are directly relevant to our research topic and target. For instance, we incorporated Yang's work, which also conducted a comparative analysis of TL and SSL methods (Yang et al., 2020).

2.3. Papers selection

As a result of predefined inclusion and exclusion criteria, this study included the research papers that met the standards. We selected 8 seminal and reviewed articles on deep learning to introduce the background of our research. We also settled on 40 articles, mainly reviews on TL and SSL, to explain the theories of the two pre-trained methods and compare their differences in the training process. In addition, we have chosen 80 transfer/self-supervised learning approaches that relate to applications in different real-world scenarios and tasks to analyse their recent advantages and limitations. All selected papers have been appropriately reviewed to develop a high-level overview to compare both pre-trained methods, from definitions to applications thoroughly.

3. Comprehensive science mapping analysis

Typically, a conventional literature review involves summarising the research content according to the perspective of the authors of the article. However, this approach may have limitations, as it relies on the author's knowledge in the specific area, potentially impacting the reliability and objectivity of the results.

In this review, we have taken a different approach. Our evaluation of selected papers primarily focuses on their backgrounds, methodologies, experiment evaluations, and conclusions. To overcome potential limitations associated with the traditional review process, we have incorporated comprehensive science mapping analysis methods using R-tool and VOSviewer. These approaches are designed to provide additional information and enhance transparency when summarising the literature's findings

3.1. Annual scientific production

The annual scientific production, as depicted in Fig. 1, serves as a metric for assessing the distribution of papers discussed in this article over the years. This metric is essential for quantifying the research focus of the article.

As mentioned above, this paper focuses on the TL and SSL approaches from 2020 to 2023, with the aim of providing an up-to-date analysis of these methods. In Fig. 1, the annual scientific production map reveals a relatively low number of citations between 2010 and 2019, with one article for the years 2010 and 2013, respectively. There are two articles for the year 2015, and one article for each year from 2016–2019. This is primarily because the studies selected from this period mostly constitute early stage theoretical research, laying the groundwork for deep learning-related research and the development of TL and SSL methods. However, there has been a notable increase in the number of articles since 2020. In 2020, the number saw a significant increase, reaching 30 articles. The trend continued in 2021, with 41 articles. Although there was a temporary drop to 17 articles in 2022, the number rebounded to 31 by 2023. These statistics align with the introductory description of this paper and reflect the comprehensive coverage of research related to TL and SSL within the past four years.

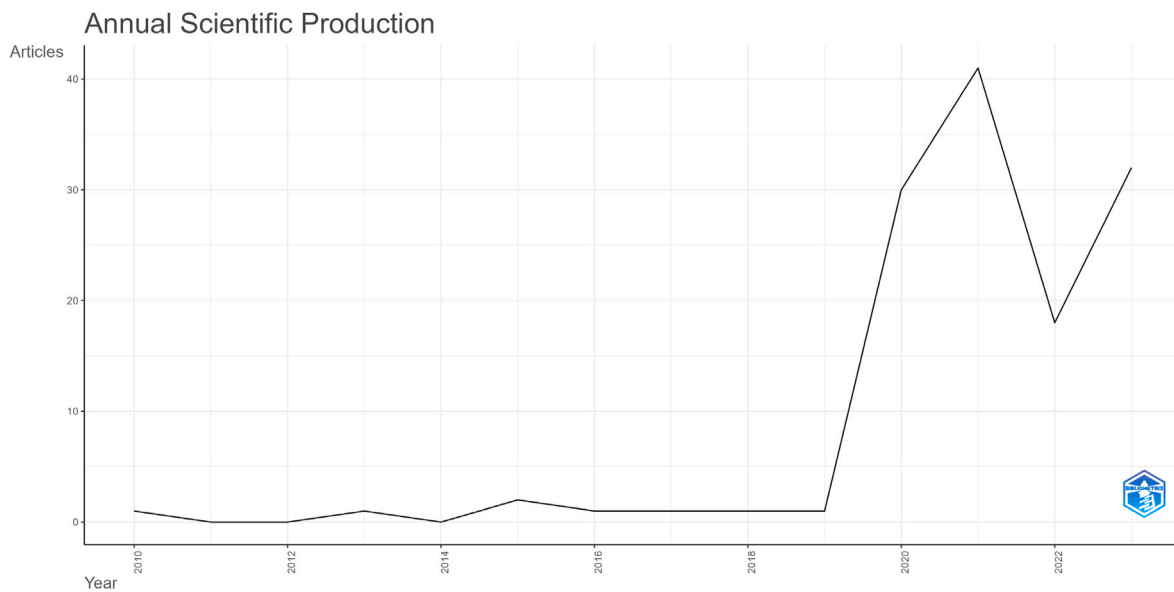


Fig. 1. Annual scientific production.

3.2. Three field plot

The three-field plot is a tool that can represent data with three parameters, allowing for the visualisation of their interrelationships. When a three-field graph is employed in the context of mapping analysis, it facilitates a more comprehensive examination of the connections between the authors, the source journals, and the specific research topics they are dedicated to. As illustrated in Fig. 2, the left field corresponds to keywords and phrases found in the title of a research paper or that closely match the title, as matched title–title (TI-TM), the middle field refers to journal sources (SO), and the right field signifies authors (AU).

As depicted in Fig. 2, the middle field (SO) shows a well-balanced distribution among various journal sources, including IEEE Transactions on Knowledge and Data Engineering & Pattern Analysis and Machine Intelligence, Journal of Big Data, Advances in Neural Information Processing Systems, and ACM International Conference, etc. Among all the journals mentioned in the plot, no single journal dominates, as the intention is to maintain diversity in the sources. Furthermore, the left field, representing keywords in titles (TI-TM), highlights the most frequently used terms, such as ‘deep’, ‘learning’, ‘transfer’, and ‘self-supervised’. These keywords, in conjunction with source journals (SO) in the middle field, underscore our review focus on deep learning-related TL and SSL research.

3.3. Word cloud

A word cloud is a tool that is used to visualise the frequency of keywords found in our reference articles. As shown in Fig. 3, it provides an overview of the keywords extracted from the reference studies. This visualisation serves to encapsulate the primary themes covered in this review and helps readers identify the central focus of this paper.

In Fig. 3, the keywords are presented in varying sizes, with larger sizes indicating more frequent mentions in the literature, and smaller sizes representing less frequent occurrences. By analysing Fig. 3, we can discern that the most prevalent topics in this article are TL, SSL, and deep learning, which align with the findings of the three-field plot. Additionally, keywords related to applications, such as classification, detection, segmentation, and representation, are frequently used. This reflects the primary focus of our paper on the application of TL and SSL methods. Other terms like ‘review’ and ‘survey’ denote our efforts to explain these pre-trained methods and compare their distinctions.

Furthermore, the keyword ‘image’ highlights the predominant data type used in this research. In summary, the word cloud illustrates that TL and SSL methods have found application across a wide spectrum of tasks, encompassing image-related, text, and audio-based applications.

3.4. Co-occurrence network

Another visualisation tool we employed in bibliometric analysis is a network of co-occurrences. This map is based on a semantic network to provide a conceptual structure for the research topics. In this section, Fig. 4 provides a co-occurrence network based on the keywords extracted from the titles of the cited articles. The network comprises nodes, the individual words in the titles, and the edges between the nodes indicate the frequency with which the words co-occur in the same title.

Fig. 4 presents a visual representation with multiple nodes, each belonging to a specific cluster identified by different node colours. Connections between these nodes indicate their proximity or associations. In Fig. 4, the nodes have been categorised into eight distinct clusters. The most frequently occurring colours in this map are red and blue, and they are closely related to TL (Transfer Learning) and SSL (Self-Supervised Learning). Within the SSL group, this cluster encompasses terms like ‘contrastive’ and ‘generative’, indicating training methods commonly used in SSL approaches. Additionally, words like ‘pose’, ‘speaker’, and ‘remote sensing’ connected with ‘estimation’ and ‘recognition’ reveal specific applications and tasks for SSL methods. Similarly, in the TL cluster, nodes contain terms such as ‘audio’, ‘image’, ‘disease’, and ‘segmentation’ which are associated with the particular applications of TL methods. Apart from these two major clusters, there are smaller clusters as well. For instance, nodes in green represent a cluster containing words like ‘network’, ‘neural’, and ‘convolutional’ that highlight the backbone structures commonly used in both TL and SSL approaches. On the contrary, the orange nodes contain terms like review, challenges, future, and applications, indicating that the primary objective of this review is to analyse the challenges posed by both TL and SSL methods and to provide information on future directions for researchers in this field.

In summary, this figure offers a visual representation of the relationships among various concepts and terms associated with deep learning and pre-trained methods, as evident from the titles of papers in this field. This visualisation provides valuable information on the current landscape of research in this area and can help identify areas where further investigation and research are warranted.

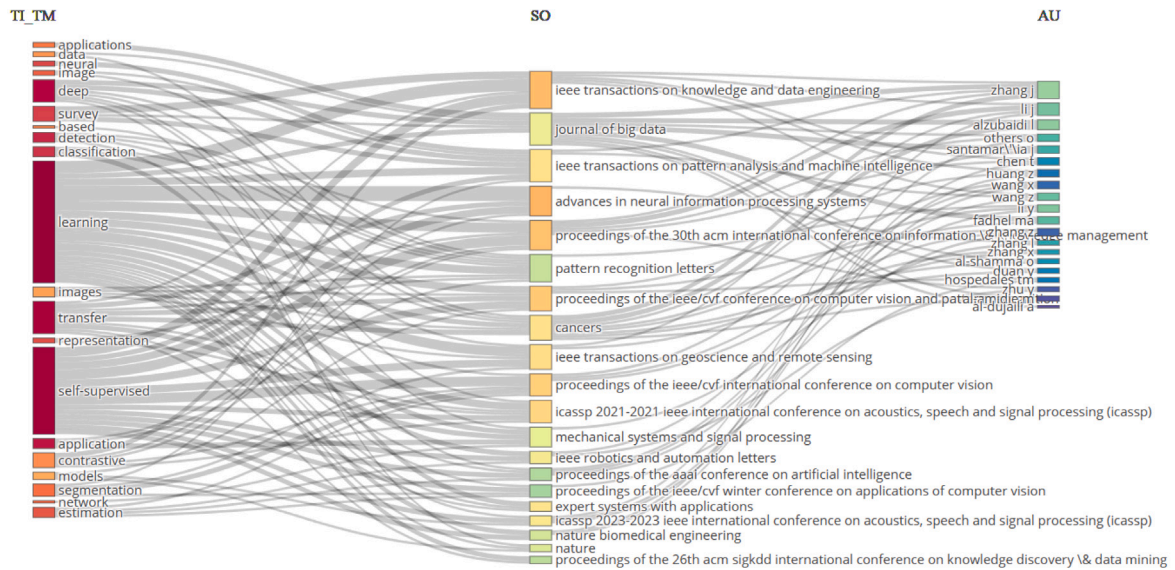


Fig. 2. Three-field plot.



Fig. 3. Word cloud.

4. Definitions for transfer and self-supervised learning

4.1. Definition of transfer learning

Transfer learning, or knowledge transfer between task domains, is a pre-training method that uses knowledge from trained data to rebuild new models (Alzubaidi et al., 2021). The study of TL firstly comes from the fact that human beings can apply previously learned knowledge wisely to solve new problems faster or provide better solutions (Pan & Yang, 2010). Traditional ML methods demonstrate effectiveness when training and testing data conform to the same feature space and distribution assumptions. However, this assumption cannot be fulfilled in many situations, which means that those models have to be rebuilt from scratch using newly collected data with a similar distribution. It is expensive and sometimes impossible to achieve in real-world scenarios. In such cases, the TL method can help reduce the need and cost of recollecting new data by transferring knowledge from a domain similar to the target domain (Alzubaidi et al., 2021).

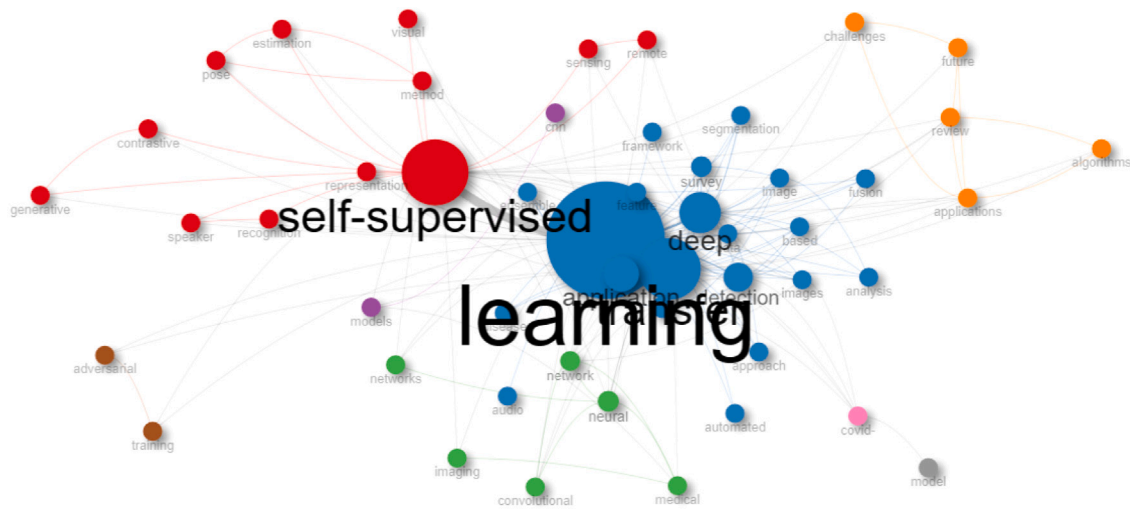
Based on Pan and Yang's very first survey paper on TL (Pan & Yang, 2010), there is a general definition of transfer learning:

Definition (Transfer Learning). Given some source domains D_{s_i} that contain a number of observations o^s , and the corresponding learning tasks τ_{s_i} (that is, $\{(D_{s_i}, \tau_{s_i}) | i = 1, 2, \dots, o^s\}$). Towards certain observations, o^t and learning tasks τ_t in the target domain D_t , the TL method aims to improve the learnt predictive function $f_t(*)$ in the domain D_t based on the knowledge it learnt from the source domain D_{s_i} and the selected learning task τ_{s_i} , where $D_{s_i} \neq D_t$ or $\tau_{s_i} \neq \tau_t$.

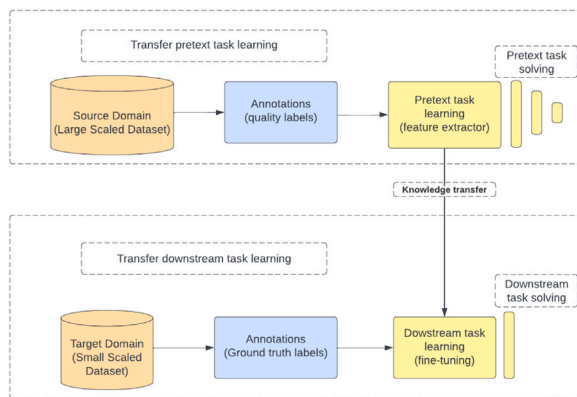
The workflow is depicted in Fig. 5. Outlines the basic process of TL, where knowledge is transferred from a domain with abundant training data to a domain lacking sufficient data to learn relevant features.

4.2. Categorisation of transfer learning

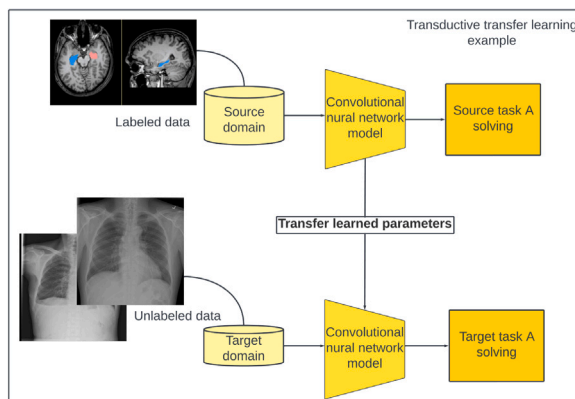
There are some different categorisation criteria for transfer learning. One standard categorisation method for TL is based on the data and



and target domains are different, but related to provide similar outer knowledge to support the learning. In the context of transductive transfer learning, the pre-trained model acquires knowledge from the source domain D_S and the source task τ_S , which is then used to improve the learning process of the target predictive function $f_T(\cdot)$, while the target domain $D_S \neq$ target domain D_T and the target task $\tau_S = \tau_T$. As shown above, Fig. 6 is an example of transductive transfer learning.



2. In the case of inductive learning, the knowledge can be transferred within the same domain. Data from the source domain can be labelled or unlabelled, while data from the target domain are usually labelled to induce an objective predictive model. In addition, based on different data situations in the source domain, inductive transfer learning can be further categorised into two cases: multi-task learning and self-taught learning. Multi-task learning happens when abundant labelled data is available in the source domain, and a small amount of labelled data exists in the target domain. Self-taught learning is usually applied when only unlabelled data are available in the source domain, and its pre-trained model uses unsupervised learning to learn knowledge and get fine-tuned with labelled data in the target domain. Further on, inductive transfer learning can be defined as follows:



With a given source domain D_S and a learning task τ_S , inductive transfer learning uses the knowledge gained from the source domain D_S and task τ_S to enhance the learning of the target predictive function $f_T(\cdot)$ in the target domain D_T , where the source and target tasks, τ_S and τ_T , respectively, are not identical.

3. Unsupervised transfer learning is performed when labelled data is unavailable in either the source domain or target domain in training. The pre-training and retraining stages used unsupervised learning frameworks, while the selected source and target tasks were different but related. Similar to the above definitions, having a source domain and learning task D_S, τ_S , and a target domain D_T with its corresponding learning task τ_T . Unsupervised transfer learning approaches enhance the target predictive function $f_T(\cdot)$ learning in task D_T with the knowledge transferred from D_S and τ_S . At the same time, the $\tau_S \neq \tau_T$ and no labels can be accessible in D_S and D_T . Fig. 8. has shown the process of an unsupervised transfer learning approach.

Another common categorisation method examines whether the source and target domains share the same feature and label spaces. They can be divided into two main categories: homogeneous and heterogeneous transfer learning.

tasks' attributes, which can be split into three main types: transductive transfer learning, inductive transfer learning, and unsupervised transfer learning.

1. The case of transductive transfer learning occurs when labelled data are available only in the source domain. In this setting, the source

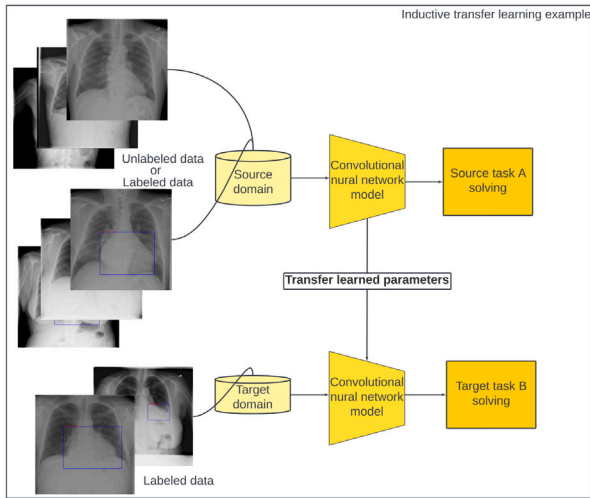


Fig. 7. Illustration of inductive transfer learning using CNN example: Two cases of inductive transfer learning use the same process in this figure, while the difference between self-taught learning and multi-task learning is based on whether data used in the training process is annotated (all CT chest images). The first neural network model follows the general TL workflow to learn from the source domain data and transfer knowledge to the target domain.

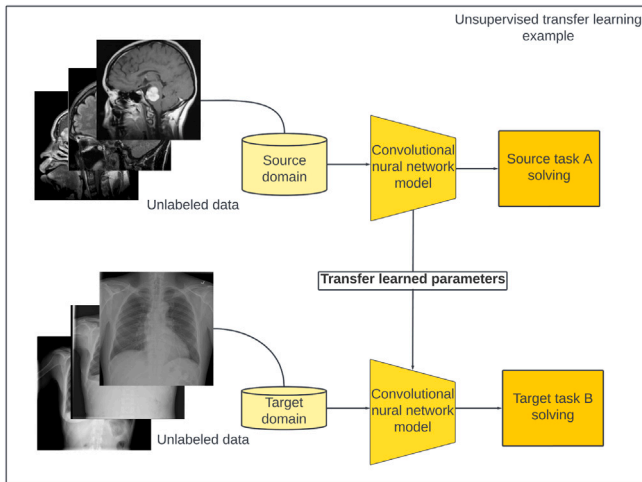


Fig. 8. Illustration of unsupervised transfer learning using CNN example: For the unsupervised transfer learning setting, the CNN model first trains with unlabelled data (brain MRI images, in this case). Later, the learned parameters from the network will be transferred to the target domain and fine-tuned with another group of unlabelled MRI images.

domain and shares the same input feature space with data from the target domain (Peirelinck et al., 2022).

2. Similar to homogeneous transfer learning, heterogeneous transfer learning is also developed for the big data environment. In this scenario, the source and target domain have their own non-overlapping feature space, which is nonequivalent and generally non-overlapping (Cody & Beling, 2023). For most heterogeneous transfer learning settings, the pre-training datasets are assumed to be drawn from the same target domain. Therefore, there are no significant distribution differences between the source and target domains. In such cases, the heterogeneous transfer learning approach needs to build a predictive model to bridge the difference in the feature spaces.

In addition to the TL settings mentioned above in this review, other interesting topics in TL are worth exploring, including transitive transfer learning (Tan, Song, Zhong, & Yang, 2015) and adversarial transfer learning (Deng, Zhang, Vodrahalli, Kawaguchi, & Zou, 2021),

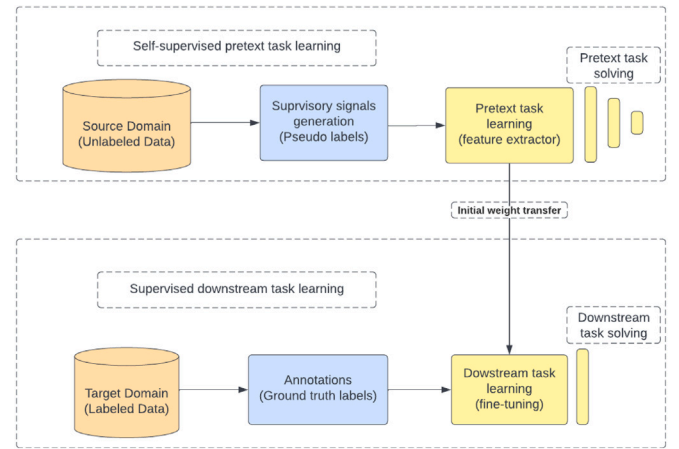


Fig. 9. Self-supervised learning main workflow.

which have gained popularity in the TL research community (Zhuang et al., 2020).

4.3. Definition of self-supervised learning

Despite the wide use of unsupervised learning in learning from the unlabelled data (Sarker, 2021), experiments have stated that the traditional unsupervised learning strategy can hardly achieve state-of-the-art performance as the fully supervised learning strategy. To improve the performance of conventional unsupervised learning approaches, self-supervised learning, a newly developed technology and a branch of unsupervised learning, has been designed to learn unlabelled data with supervision under self-generated pseudo labels. Facing the general lack of annotated data in many fields such as medical, satellite photographs, and video sources, SSL approaches have proved their effectiveness in learning powerful representations from unlabelled data, improving the model's robustness, and against data noises (Chen, Fu, Liu, Li and Lee, 2023; Ericsson, Gouk, & Hospedales, 2021; Hochberg, Greenspan, & Giryes, 2022; Rani, Nabi, Kumar, Mittal, & Kumar, 2023). In the pre-training stage, pseudo-labels are generated using the supervisory signals from unlabelled data. This enables the model to capture the semantic difference without the need for human annotation (Liu et al., 2021). Generally, the SSL approach uses two types of tasks to build up its learning pipeline: pretext and downstream tasks. An SSL workflow is shown in Fig. 9. as follows.

In summary from Lee and Lei's article (Lee, Lei, Saunshi, & Zhuo, 2021), a definition of SSL is shown as follows:

Definition(Self-supervised learning): Given a source domain D_s , which contains large numbers of unlabelled observations O^u and a target domain D_t with small numbers of labelled observations O^l . SSL learns from unlabelled observations O^u to generate pseudo labels P_l and is trained to solve pretext auxiliary tasks τ_{s_i} to extract effective representations. The learned knowledge is transferred to the target domain as initial weights, fine-tuned with the labelled observations O^l , and supervised downstream task τ_{t_i} learning.

4.4. Categorisations of self-supervised learning

4.4.1. Pretext tasks

Pretext tasks, referred to as auxiliary or pre-training tasks, serve as intermediate tasks within machine learning. In SSL, a feature extractor is trained to accomplish a pretext task using unlabelled data. This pretext task serves as an additional source of training signal, guiding the model to learn valuable representations or features that can later be transferred to the target task (Albelwi, 2022). In the pre-training stage, an encoder is used to learn knowledge from unlabelled data in a

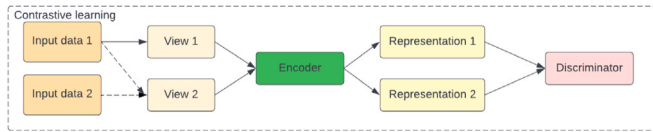


Fig. 10. Contrastive learning workflow.

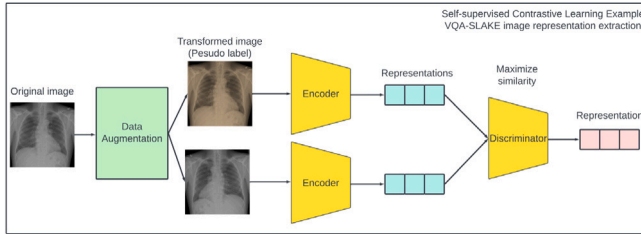


Fig. 11. Illustration of contrastive learning using encoder example: A raw unlabelled CT chest image is selected as the input data. The following data pair is generated using random augmentation operations, with two encoders applied separately to extract their corresponding representations. Finally, a discriminator will be used to maximise the representation's similarity to conclude a common representation.

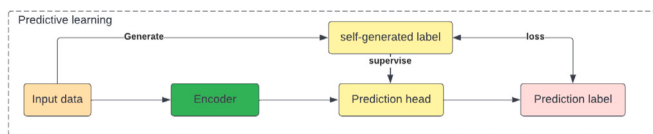


Fig. 12. Predictive learning workflow.

supervised fashion, where the labels are generated from the data itself. Pretext task learning is the most significant component in SSL because it decides what knowledge the pre-trained model learns from the data. As Ericsson, Gouk, Loy, and Hospedales (2022) stated, what defines SSL is its pretext task. Based on typical learning methods, the pretext tasks are usually categorised into three groups: contrastive, predictive, and generative learning (Wu, Lin, Tan, Gao, & Li, 2021).

1. Contrastive learning is a representation learning approach that aims to maximise the agreement between two positively paired samples that are jointly selected while minimising the agreement between two negatively paired samples chosen independently. Positive examples are obtained by subjecting an input image to random augmentations, yielding two transformed views of the same image. Negative examples, on the other hand, comprise any images that differ from these transformed views. Given sample data-data pairs, the selected contrastive encoders will learn from input data to compute the representations of views and then use a discriminator module to compare the similarity of instances and calculate the contrastive loss (Tian et al., 2020). Fig. 10. Illustrates the workflow for contrastive learning as follows:

As a trivial example, a pseudo-label can be generated by transforming the input image into other states using data augmentation approaches. For the following case in Fig. 11., the pseudo-labels are generated by colour-changing and flipping.

2. Predictive learning uses data-label pairs to train the encoders under the supervision of informative labels (Xie, Xu, Zhang, Wang, & Ji, 2022). The required training labels are self-generated from data, making predictive learning lie under a part of SSL. After picking up specific properties or parts from the input data, the predictive models will predict a new label using the trained prediction head. Ultimately, the system compares the actual and predicted labels to provide a reasonable loss. Fig. 12 is a simple workflow for predictive learning.

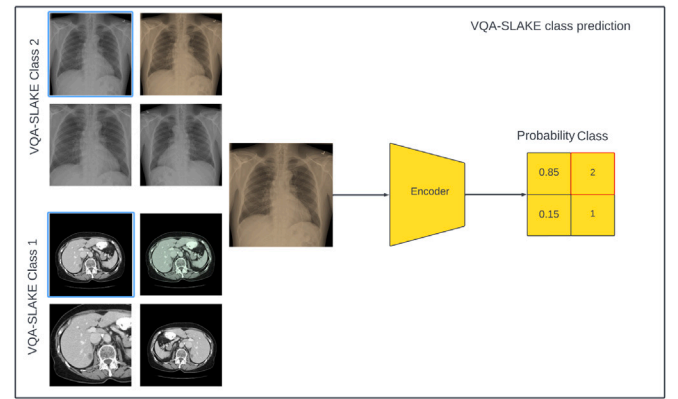


Fig. 13. Illustration of predictive learning using encoder example: The marked images in blue exemplary patches are the raw unlabelled data extracted from the VQA-SLAKE dataset. The remaining generated images are grouped to compose a small image class to the raw data. Later, an encoder model is applied to learn feature representation from image sets by predicting the correct class for the selected image.

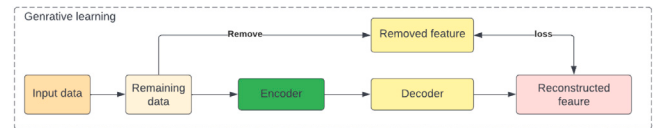


Fig. 14. Generative learning workflow.

Another practical example of predictive learning is shown in Fig. 13. The pseudo labels used augmentation techniques such as colour-changing, flipping, and enlarging. The transformed images help expand the raw data into two distinct classes and provide representations for the encoder to learn from.

3. Generative learning aims to capture inherent relationships within data by reconstructing the original input using a corrupted version (Tao, Qi, Guo, Zhu, & Li, 2023). The normal operation step for the generative model is to mask parts of the input data first and use self-supervision to acquire the embeddings of information from the input data and reconstruct the missing pieces of the data. As shown in Fig. 14., the model compares the reconstructed and original features to calculate the loss.

A generative learning example is Fig. 15 aims to solve the data denoise task. The pseudo label here is generated from a generator by adding noise distribution to the original image, and the encoder learns the estimated distributions from the pseudo label to generate a more precise image.

4.4.2. Downstream tasks

The downstream task refers to the ultimate objective that the SSL model aims to fine-tune after completing pre-training on a source task. These downstream tasks can be classified based on the learning objectives specific to the target domain and the characteristics of the training data. This review focuses exclusively on downstream tasks related to widely used data, including node, link, and graph level tasks.

Node-level Tasks

Node-level tasks focus on the properties of the nodes and are often used for classification purposes. With several nodes corresponding to several labels, the graph encoder will be used to learn from the labels and the map between labels and nodes to build node embeddings and make predictions about the labels of unlabelled data.

Link-level Tasks

Link-level tasks pay more attention to learning the representations of node pairs and edge properties. To learn edge properties, the model

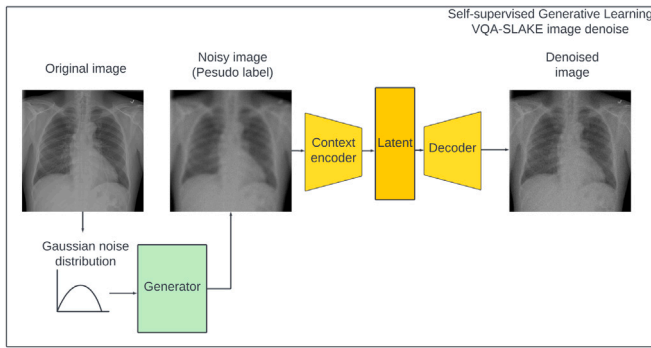


Fig. 15. Illustration of generative learning using auto-encoder example: A noisy image has been generated by adding Gaussian noise to the original image. An auto-encoder is deployed to learn representations by first using its encoder network to compress a noisy image into a latent space, followed by its decoder network to reconstruct the compressed image from the same latent space.

will be trained to determine the presence or absence of an edge between two given nodes.

Graph-level Tasks

Graph-level tasks involve training on multiple graphs within a dataset to predict specific properties or characteristics of individual graphs. For given graphs with corresponding properties that are known, one graph encoder is trained on labelled data to learn graph-level representations and then predict the unknown attributes.

4.4.3. Training strategies

The training strategies in SSL can be categorised into three categories: Pre-training and Fine-tuning (P&F), Joint Learning (JL), and Unsupervised Representation Learning (URL).

Pre-training and Fine-tuning (P&F)

In the P&F strategy, a two-stage paradigm is used to train the model. The encoder f_θ target domain's learning objectives and the training data's characteristics θ as the initialisations for the new encoder $f_{\theta_{new}}(\cdot)$. During the fine-tuning stage, the pre-trained encoder $f_{\theta_{new}}(\cdot)$ is further trained by fine-tuning its parameters. This is done with a prediction head $g_{new}(\cdot)$, specifically designed for downstream tasks.

Joint Learning (JL)

In this approach, the encoder $f_\theta(\cdot)$ and the prediction head $g(\cdot)$ are trained together using both pretext tasks and downstream tasks for supervision. This joint learning strategy can be considered as a form of multitask learning, where the pretext tasks act as a regularisation mechanism. By combining the training signals from both types of tasks, the model can effectively learn representations beneficial for both the pretext tasks and the downstream tasks.

Unsupervised Representation Learning (URL)

This scheme follows a two-stage training paradigm, where the encoder is initially pre-trained with the selected pretext tasks providing supervision. Then, the pre-trained parameters θ are kept frozen in the second stage, and the model is trained using only downstream tasks on these frozen representations. On the other hand, the URL strategy shares similarities with the P&F strategy, as they both initialise the downstream task encoder with the parameters from the pre-trained encoder. However, the URL strategy faces a more challenging training process since its pre-trained encoder is optimised solely during the pre-training and remains frozen in the second stage. In contrast, the pre-trained encoder in the P&F strategy can undergo further optimisation through fine-tuning with the supervision of the downstream task.

Table 1

Comparison of Transfer learning and Self-supervised learning.

Elements	Transfer learning	Self-supervised learning
Training strategy		
Supervised learning	✓	✓
Unsupervised learning	✓	✗
Self-Supervised learning	✗	✓
Transfer approach		
Feature -Representation transfer	✓	✓
Instance transfer	✓	✗
Parameter Transfer	✓	✓
Relational transfer	✓	✓
Intra-Data information transfer	✗	✓
Training data		
Labelled data	✓	✓
Unlabelled data	✓	✓
Training tasks		
Same source -Target task	✓	✗
Different source -Target task	✓	✓

5. Comparison of transfer learning and self-supervised learning

This section examined four key components of the TL and SSL training process: training strategy, transfer approach, training data, and training task relationship. TL and SSL generally share a similar training process since they use a two-stage training paradigm, learn features from one domain, and transfer them to another. The main difference between them is that TL uses supervised or unsupervised learning strategies that rely on manually annotated labels to learn representations from data. At the same time, SSL trains data under supervision using self-generated labels. Furthermore, other similarities and differences between the two methods are summarised in Table 1, as shown below.

5.1. Comparison of training strategy

Different training strategies can be employed at each stage of the training process for pre-training methods such as TL and SSL. There is no fixed regulation for the choice of training strategy, and it usually depends on the nature of the source and target domains, the data environment, and the task requirements.

Based on the explanations of the TL settings we listed in the previous section, a TL approach can use both supervised and unsupervised learning strategies during the pre-training and retraining stages. Supervised learning aims to solve a target problem by building a predictive model to learn strong feature representations from a labelled dataset. Deep learning is a typical supervised learning method, which always combines the predictive model with a representation extractor function and a classifier or regression function. Unlike supervised learning, unsupervised learning techniques can learn from unlabelled data through generative models and density estimators. Some standard unsupervised learning methods, including Gaussian mixtures, Variational Auto-Encoder (VAE) (Kingma & Welling, 2013), Clustering, and Generative

Adversarial Network (GAN) (Goodfellow et al., 2020) can be trained to make use of the input data's compact feature to reconstruct the missing data. To summarise, supervised learning methods tend to make estimates from previous data, and unsupervised learning approaches focus more on restoring existing data. Besides, labels can help supervised learning learn more robust features than unsupervised learning but cost more human resources and expense simultaneously.

In comparison, the two SSL training stages are all under supervision. In the pre-training stage, the approach uses self-generated pseudo-labels to supervise the learning process. In the retraining scene, the approach uses labels from raw data to conduct supervised learning and keep the learning process under supervision.

5.2. Comparison of transfer approach

The primary method for TL and SSL is to extract knowledge from one domain and transfer it to the target domain tasks. TL and SSL methods typically acquire distinct types of learning from the source data, as they are designed to meet specific target task requirements.

For the TL method, the approaches are generally designed to learn representative knowledge from training data to avoid the lack of expert knowledge from labels. The experiment also indicates that feature reuse and low-level data statistics play essential roles in TL, and the quality of transferred features can significantly impact the overall performance (Neyshabur, Sedghi, & Zhang, 2020). Based on previous studies (Iman, Arabnia, & Rasheed, 2023; Pinto, Wang, Roy, Hong, & Capozzoli, 2022), there are four common approaches of knowledge transferring: instance transfer, feature representation transfer, parameter transfer, and relational knowledge transfer, which corresponds to the respective knowledge that is used in the TL training process.

- Feature-representation-based transfer approach aims to identify compelling features or representations to represent the data characteristics, reduce domain divergence and enhance model performance in classification and regression tasks.
- Parameter-transfer approaches share parameters or prior distributions of hyperparameters learned from the source domain with the target domain to promote efficient learning.
- Instance-transfer approaches are motivated by importance sampling, designed to utilise and reweight a part of the source data to enlarge the limited data from the target domain. Consider the fact that the source domain data cannot be directly reused.
- Relational-knowledge-transfer approaches focus on relationship knowledge transfer between relational domains, where data are related and do not follow an identical distribution.

In contrast, SSL approaches are different from TL methods since the knowledge learning is motivated by the pretext tasks (Jing & Tian, 2020). The type of retrieval representative elements is based on the pretext task settings.

- Contrastive learning is a simple approach that employs a similarity metric to distinguish and separate similar and dissimilar samples. The model extracts general feature representations that can be transferred to downstream tasks by clustering samples. Thus, contrastive learning can be viewed as a feature-representation transfer approach.
- Generative learning approaches focus more on sharing parameter and intra-data information. The everyday tasks used to learn knowledge in this setting include generating new data, predicting and reconstructing the missing part of the original data, etc. To achieve these tasks, generative learning approaches utilise attribute and spatial structure information as self-supervision signals, which can be transferred to the target domain (Wu et al., 2021). Furthermore, these methods can use learned parameters to establish a foundation and create rich information embedding, serving as natural self-supervision for learning out-of-distribution features (Mohseni, Pitale, Yadawa, & Wang, 2020).
- Predictive learning aims to handle the relationships between data and labels. In node-level predictive tasks, the node properties have been used as labels to provide supervision. The local and global contextual

information would be extracted as labels in context-based predictive tasks. Also, in some tasks that lack specific relational information, the model can generate pseudo labels by clustering data from the previous stage or randomly assigning data as labels. The main idea for predictive learning approaches is based on the relational information between data, which is similar to the relational transfer approach in the TL method.

Moreover, in generative learning, specific node-level pretext tasks can also acquire relational information from node connections. Nonetheless, these methods for learning node relations are founded on intra-data information instead of input data, distinguishing them from the relational knowledge transfer approach employed in TL.

5.3. Comparison of training data

Different from the traditional supervised or unsupervised learning approaches that focus on handling specific data types. TL and SSL methods have taken advantage of both of the data. Among three standard TL settings, transductive transfer learning trains annotated data at the pre-training stage and fine-tuned with unlabelled data at the retraining stage. Inductive transfer learning also uses labelled and unlabelled data during training. In contrast, unsupervised transfer learning differs because it only uses unlabelled data in the training stages (Niu, Liu, Wang, & Song, 2020).

On the other hand, despite the whole training process of SSL being under supervision, the data used in pretext task training are unlabelled, with only a small amount of labelled data for downstream task training. In summary, TL uses labelled or unlabelled data at each training stage, while SSL uses unlabelled data in pre-training and labelled data in retraining.

5.4. Comparison of training tasks

Training task has played a crucial role in the learning model, guiding feature learning and representation generalisation. Referring to Pan's review (Pan & Yang, 2010), different TL settings may cause variations in the relationships between the source and target tasks. The source and target domains share the same task in a transductive transfer learning setting. The samples used to design tasks in the target domain must be carefully selected to achieve optimal performance in the target domain. Meanwhile, the tasks in the source and target domains are different in the inductive transfer learning setting, considering that inductive transfer learning is within the same domain. The source domain tasks aim to use the training data to induce the target predictive function. In unsupervised transfer learning, approaches favour different but related tasks for the source and target domains. The selection of tasks aims to enable the model to learn common representations across disciplines, even in scenarios where only a few or no labelled data are available in the domain.

At the same time, the relationship between SSL's pretext training tasks and downstream tasks is weak. The pretext tasks in SSL are hand-crafted auxiliary tasks that learn with pseudo labels, while downstream tasks are designed based on manual labels (Liu et al., 2022). The pretext task aims to learn generalised representation and benefit the downstream task with better initialisation.

6. Latest applications of transfer learning and self-supervised learning

In this section, we analyse recent applications of TL and SSL methods that have achieved state-of-the-art performance in specific evaluation metrics. Table 2 presents the articles classified into eleven main groups depending on their specific application areas. Based on the table, the "Methods" column indicates the training methods used in each research, the column "Articles" gives reference to the corresponding articles, and the "Application Tasks" describes the downstream tasks

Table 2
Summary of recent TL and SSL applications.

No.	Methods	Articles	Application tasks	Categories	Architectures	Performance (%)
1	TL	Xue et al. (2020)	Cervical histopathology	Medical/Image classification	Ensembled TL	98.61/Acc
2	TL	Lopes et al. (2021)	Heart disease detection	Medical/Image classification	Sex identify model	83/Acc
3	TL	De Bois, El Yacoubi, and Ammi (2021)	Blood glucose prediction	Medical/Image classification	Adversarial FCN	18.94/RMSE*
4	SSL	Fedorov et al. (2021)	Alzheimer prediction	Medical/Image classification	DCGAN	95/AUC
5	TL	Bargshady et al. (2022)	COVID-19 detection	Medical/Image classification	Inception-CycleGAN	94.2/Acc
6	TL	Gardner, Bull, Dervilis, and Worden (2022)	Structural monitoring	Medical/Image classification	KBTL	98.5/Acc
7	TL	Kathamuthu et al. (2023)	COVID-19 detection	Medical/Image classification	VGG-16	98/Acc
8	SSL	Jiao, Droste, Drukker, Papageorgiou, and Noble (2020)	Ultrasound video	Medical/Video classification	Siamese network	75.7/F1
9	SSL	Xue and Salim (2021)	COVID-19 classification	Medical/Audio classification	Transformer-CP	84.43/Acc
10	TL	McCreery, Katariya, Kannan, Chablani, and Amatriain (2020)	COVID-19 FAQs	Medical/Text generation	BERT-QA	84.5/Acc
11	SSL	Yoon, Zhang, Jordon, and van der Schaar (2020)	Patient treatment prediction	Medical/Tabular prediction	VIME	86.02/AUC
12	TL	Ankenbrand et al. (2021)	MRI segmentation	Medical/Image segmentation	Modified TL	90/DSC
13	TL	Zhao et al. (2021)	Fundus segmentation	Medical/Image segmentation	TL based U-Net	99.75/Acc
14	SSL	Azizi et al. (2023)	Medical vision	Medical/Multi-task	REMEDIS	95.8/AUC
15	SSL	Moor et al. (2023)	Generalist medical AI	Medical/Multi-task	GMAI	–
16	TL	Zhang et al. (2020)	Ball screw diagnosis	Industrial/Image classification	IEDT	94.95/Acc
17	TL	Xin, Cheng, Diender, and Veljkovic (2020)	Bridge engineering	Industrial/Image classification	TL-based CNN	99.05/Acc
18	TL	Zhu, Chen, Anduv, Jin, and Du (2021)	Chillers diagnosis	Industrial/Image classification	FDD based TL	81.27/Acc
19	TL	Michau and Fink (2021)	Anomaly detector	Industrial/Image classification	Adversarial TL	99.8/Acc
20	TL	Asanuma, Doi, and Igarashi (2020)	Electric motor	Industrial/Regression	VGG16	–
21	SSL	Ren, Wang, Laili, and Zhang (2021)	Soft sensor	Industrial/Regression	LSTM-DeepFM	0.7479/RMSE*
22	SSL	Akrim, Gogu, Vingerhoeds, and Salaün (2023)	Fatigue damage prediction	Industrial/Regression	SSL-based	13.24/MAPE
23	TL	Zhang, Wang, and Li (2023)	Fault diagnosis	Industrial/Federated learning	Blockchain framework	99.8/Acc
24	SSL	Yuan and Lin (2020)	City classification	Satellite/Image classification	SITS-BERT	98.76/Acc
25	SSL	Li et al. (2023)	Scene classification	Satellite/Image classification	MES ² L	95.03/Acc
26	SSL	Muhtar, Zhang, and Xiao (2022)	Object segmentation	Satellite/Image segmentation	IndexNet	71.07/mIoU
27	TL	Chen, Sun, Li, and Hou (2022)	Plane detection	Satellite/Object detection	DA R-CNN	90.17/AP
28	TL	Gomroki, Hasanlou, and Reinartz (2023)	City change detection	Satellite/Object detection	EfficientNetV2 T-Unet	97.66/Acc
29	SSL	Xiao et al. (2023)	Image super-resolution	Satellite/Image degradation	SSL network	–
30	SSL	Liu, Jiang, Xiong, Yang, and Ye (2020)	Chat bot	NLP/Text generation	MRTM	87.7/F1
31	SSL	Cao, Jin, Wan, and Yu (2020)	Auto essay scoring	NLP/Text generation	HA-LSTM+SST+DAT	79.7/QWK
32	TL	Van Nguyen, Nguyen, Min, and Nguyen (2021)	Crosslingual translation	NLP/Text translation	CCCAR	72.1/F1
33	TL	Omran, Sharef, Grosan, and Li (2023)	Crosslingual translation	NLP/Text translation	LSTM model	96.65/F1
34	TL	Lu et al. (2021)	Event extraction	NLP/Information retrieval	TEXT2EVENT	72.7/F1
35	TL	Elmadany, Abdul-Mageed, et al. (2022)	Crosslingual task	NLP/Multi-task	AraT5	24.37/Bleu
36	TL	Xu, Dinkel, Wu, Xie, and Yu (2021)	Audio caption generation	NLP/Multi-modal	CNN10	65.5/Bleu
37	TL	Sung, Cho, and Bansal (2022)	Vision-text generation	NLP/Multi-modal	CLIP-BART	79.2/Acc
38	SSL	Li et al. (2020)	3D pose estimation	Motion/Regression	Modified SSL	41.4/AUC
39	SSL	Bhatnagar, Sminchisescu, Theobalt, and Pons-Moll (2020)	3D pose estimation	Motion/Regression	LoopReg	–
40	SSL	Dai et al. (2020)	3D depth estimation	Motion/Regression	Depth-net	72.31/IoU
41	SSL	Spurr, Dahiya, Wang, Zhang, and Hilliges (2021)	Hand pose estimation	Motion/Regression	PeCLR	74/AUC
42	SSL	Cheng, Chen, Zhang, and Lin (2021)	Action recognition	Motion/Regression	Motion-Transformer	91.9/Acc
43	SSL	Ma, Li and Li (2023)	3D pose estimation	Motion/regression	Factorisation network	40.4/AUC
44	TL	Huang, Fu, Liu and Ostadabbas (2021)	Infant pose estimation	Motion/Image estimation	FIDIP	93.6/mAP
45	SSL	Baur et al. (2021)	3D scene segmentation	Motion/Image segmentation	SLIM	0.0668/AEE*
46	SSL	Huang et al. (2021)	Action recognition	Motion/Video classification	MoSI	71.8/Acc
47	SSL	Bi, Hu, Zhao, Li, and Sun (2023)	Action recognition	Motion/Video classification	Contrastive-SSL	75.7/Acc
48	TL	Cao and Xiang (2020)	Garbage classification	Environment/Image classification	Modified Inception-V3	99.3/Acc
49	TL	Choe, Choi, and Kim (2020)	Parrot classification	Environment/Image classification	Keras-based TL	96.75/F1
50	TL	Rehman et al. (2021)	Leaf disease recognition	Environment/Image classification	ResNet+Mask RCNN	96.6/Acc
51	TL	Huang, Chuang, and Liao (2022)	Tomato pest identification	Environment/Image classification	ResNet50+DA	97.12/Acc
52	TL	Attallah (2023)	Tomato leaf classification	Environment/Image classification	Resnet-18	99.34/Acc
53	TL	Palanisamy, Singhanian, and Yao (2020)	Environment sound classification	Environment/Audio classification	Ensemble DenseNet	92.89/Acc
54	TL	Chen et al. (2021)	Water quality prediction	Environment/Text prediction	TrAdaBoost-LSTM	0.47/RMSE*
55	SSL	Shen et al. (2020)	Taxonomy expansion	Web/Information retrieval	TaxoExpan-FWFS	72.3/F1
56	SSL	Yao et al. (2021)	Item recommendation	Web/Information retrieval	SSL-DNN	53.55/Recall
57	SSL	Zhang et al. (2021)	Group recommendation	Web/Information retrieval	S ² -HHGR	88.3/Recall
58	SSL	Zhou, Dou, Zhu, and Wen (2021)	Personalised search	Web/Information retrieval	PSSL	83.01/mAP
59	SSL	Feng, Wan, Wang, Li, and Luo (2021)	Twitter bot account	Web/Text classification	SATAR	95.09/Acc
60	TL	Ameer et al. (2023)	Emotion classification	Web/Text classification	RoBERTa-MA	62.4/Acc
61	TL	Arbane, Benlamri, Brik, and Alahmar (2023)	Emotion classification	Web/Text classification	Bi-LSTM	93/Recall

(continued on next page)

Table 2 (continued).

No.	Methods	Articles	Application tasks	Categories	Architectures	Performance (%)
62	SSL	Huh, Heo, Kang, Watanabe, and Chung (2020)	Speaker recognition	Speaker/Audio classification	Light ResNet-34	8.65/EER
63	SSL	Chi et al. (2021)	Speaker classification	Speaker/Audio classification	Audio ALBERT	99.3/Acc
64	SSL	Xia, Zhang, Weng, Yu, and Yu (2021)	Speaker verification	Speaker/Audio classification	MoCo+WavAug	8.63/EER
65	SSL	Sang, Li, Liu, Arnold, and Wan (2022)	Speaker verification	Speaker/Audio classification	Thin-ResNet34	6.99/EER
66	SSL	Chen et al. (2023)	Speaker verification	Speaker/Audio classification	WavLM-based SS	16.5/WER
67	SSL	Cai, Wang, and Li (2021)	Speaker recognition	Speaker/Audio segmentation	Iterative SSL	3.45/EER
68	SSL	Huang, Raj, Garcia, and Khudanpur (2023)	Speaker recognition	Speaker/Object localisation	JSM	7.58/WER
69	SSL	Hu et al. (2020)	Speaker localisation	Speaker/Multi-modal	Modified SSL	51.9/IoU
70	SSL	Song, Wang, Fan, Tan, and Zhang (2022)	Speaker localisation	Speaker/Multi-modal	SSPL	61/AUC
71	SSL	Li, Huang, and Zhang (2021)	Vehicle re-identify	Vehicle/Image classification	SSL network	98.7/Acc
72	SSL	Kothandaraman, Chandra, and Manocha (2021)	Road segmentation	Vehicle/Image segmentation	SS-SFDA	95.63/F1
73	SSL	Kumar et al. (2021)	Road segmentation	Vehicle/Image segmentation	SynDistNet	1.668/RSMSE*
74	TL	Hu, Zhao and Zhang (2020)	Infrared pedestrian detection	Vehicle/Object detection	Faster R-CNN	84.78/AP
75	TL	Farid et al. (2023)	Vehicle detection	Vehicle/Object detection	YOLO-v5	99.94/Acc
76	SSL	Wei et al. (2023)	Depth estimation	Vehicle/Image prediction	Transformer	6.835/RMSE*
77	SSL	Yan, Zhu, Jin, and Bohg (2020)	Robot manipulation	Robot/Image prediction	Dynamic model	–
78	SSL	Nubert, Khattak, and Hutter (2021)	LiDAR localisation	Robot/Image estimation	Modified SSL	–
79	SSL	Ze, Hansen, Chen, Jain, and Wang (2023)	Motor control	Robot/Image reconstruction	3D auto-encoder	–
80	TL	Inubushi and Goto (2020)	Fluid turbulence prediction	Physics/Nonlinear prediction	Modified TL	0.1/MSE

* Some evaluation metrics use their original performance number without being represented as a percentage.

in the real world they focus on. The “Categories” contain the groups to which they belong and are followed by the general data and task types they are applying for. The “Architectures” column presents the framework proposed in the study, while the “Performance” section displays the highest performance achieved by these architectures in their respective evaluation metrics.

6.1. Related evaluation metrics

In Table 2, various evaluation methods have been utilised to assess the performance of the studies. In this part, we will present and explain the evaluation methods used in the applications for the sake of clarity and comparison.

1. Accuracy

Accuracy is a straightforward evaluation metric that measures the proportion of correct model predictions out of all predictions made. A higher accuracy score indicates that the model is making more accurate predictions. The formula for accuracy is as follows: (TrueNeg represents the number of accurate negative predictions, and TruePosi represents the number of true positive predictions.)

$$Accuracy = \frac{TrueNegatives + TruePositive}{TrueNeg + FalseNeg + TruePosi + FalsePosi}$$

2. Precision

Precision is a performance metric that focuses solely on the model's ability to predict positive class. A higher precision score indicates that the model makes fewer mistakes in positive predictions. The precision formula is as follows:

$$Precision = \frac{TruePositive}{TruePositive + FalsePositive}$$

3. Recall

The recall is an evaluation metric measuring the percentage of actual positive instances a model correctly identified as positive (True Positive). A higher recall indicates that the model makes fewer errors in predicting true positive classes. The formula for the recall is as follows:

$$Recall = \frac{TruePositive}{TruePositive + FalseNegative}$$

4. F1 score

The F1 score is a metric that combines precision and recall, calculated as their harmonic mean. It provides a balanced evaluation of both

metrics and considers the number and type of errors the model makes. A higher F1 score indicates higher precision and recall. The formula is as follows:

$$F1 = 2 * \frac{Precision * Recall}{Precision + Recall}$$

5. (Mean) average precision (AP/mAP)

Mean average precision (mAP) is a metric commonly used to evaluate the performance of object detection and segmentation systems. It is calculated as the average of each class's average precision (AP) values, with AP calculated over recall values ranging from 0 to 1. The formula for mAP is as follows:

$$mAP = \frac{1}{n} \sum_{k=1}^{k=n} AP_k$$

6. ROC curve (AUC)

A ROC curve is a graphical representation of the performance of a classification model at various threshold settings. It plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at different classification thresholds. TPR is the proportion of actual positive cases that are correctly identified as positive by the model. At the same time, FPR is the proportion of real negative cases that are incorrectly identified as positive by the model.

$$TPR = \frac{TruePositive}{TruePositive + FalseNegative}$$

$$FPR = \frac{FalsePositive}{FalsePositive + TrueNegative}$$

The area under the ROC Curve (AUC) measures the degree of separability, indicating the model's ability to distinguish between different classes in a classification problem. A higher AUC score indicates better performance by the model in correctly predicting the classes. The following is an example of a ROC-AUC curve: (see Fig. 16).

7. (Root) Mean squared error (MSE/RMSE)

The Mean Square Error (MSE) is a measure that computes the average of the squared differences between the predicted and observed values. A lower MSE implies that the model's predictions are closer to the ground truth, indicating better performance. Root Mean Square Error (RMSE) is the square root of the MSE, and it represents the normalised distance between the predicted and actual value vectors. The formulas for MSE and RMSE are shown below:

$$MSE = \frac{1}{n} \sum_{k=1}^n (Y_k - \hat{Y}_k)^2$$

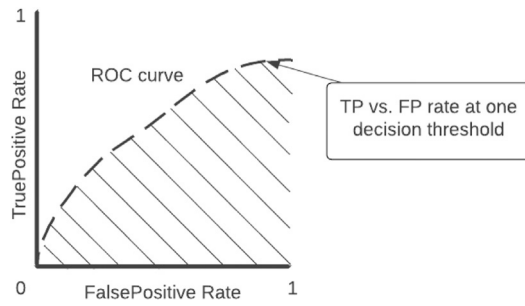


Fig. 16. ROC-AUC curve sample.

$$RMSE = \sqrt{\sum_{k=1}^n \frac{(Y_k - \hat{Y}_k)^2}{n}}$$

8. Mean absolute percentage error (MAPE)

The mean absolute percentage error (MAPE) is a performance metric utilised to evaluate the precision of forecasted quantities relative to actual quantities. It determines the average of the absolute percentage errors for each entry within a dataset. MAPE is especially valuable when analysing extensive datasets and necessitates the inclusion of nonzero values. The formulas below illustrate the computation of MAPE:

$$MAPE = \frac{1}{n} \sum_{k=1}^n \left| \frac{Y_k - \hat{Y}_k}{Y_k} \right| * 100$$

9. Dice coefficient (DSC)

The Dice coefficient (DSC) is a statistical metric that measures the similarity between two samples. A higher DSC value indicates a better performance of ITL models. The formula for DSC is given below:

$$DSC = \frac{2TruePositive}{2TruePositive + FalsePositive + FalseNegative}$$

10. Average endpoint error (AEE)

Endpoint error (EPE) is a statistical method that measures the Euclidean distance between an estimated optical flow vector V_{es} and its corresponding ground-truth optical flow vector V_{gr} . Average endpoint error (AEE) is the mean Euclidean distance between a set of estimated optical flow vectors and their corresponding ground truth vectors. A lower AEE indicates better model performance. The formula for AEE is as follows:

$$AEE = \frac{1}{N} \sum_x \|V_{es}(X) - V_{gr}(X)\|_2$$

11. Equal error rate (EER)

The Equal Error Rate (EER) is a statistical method that measures the performance of a verification task. It is the point on a ROC or DET curve where the false acceptance and rejection rates are equal. The lower the EER, the more accurate the verification model.

12. Intersection over union (IoU)

The Intersection over Union (IoU) metric is commonly used to assess the performance of object detection tasks. It measures the extent to which the predicted boundary aligns with the actual boundary of the object. A higher IoU score indicates better object detection accuracy, reflecting a more significant overlap between the predicted and ground-truth bounding boxes.

13. Bilingual evaluation understudy score (Bleu)

The BLEU score serves as an evaluation metric for automatic machine translation systems, assessing the quality of generated sentences compared to their reference sentences. It quantifies the similarity between the two sets of sentences by computing a score. A higher score

signifies a stronger alignment, with a perfect match achieving a score of 1.0, while a lack of alignment corresponds to a score of 0.0.

14. Quadratic weighted kappa (QWK)

The quadratic weighted kappa (QWK) is a metric used to evaluate the level of agreement between two annotators. The QWK score ranges from -1 to 1, with negative values indicating that the model performs worse than random and a score of 1.0 indicating perfect agreement.

15. Word error rate (WER)

Word error rate (WER) serves as a prevalent metric to assess the performance of speech recognition or machine translation systems. It is derived from the Levenshtein distance but operates at the word level instead of the phoneme level. The WER can be computed using the formula:

$$WER = \frac{S + D + I}{S + D + C}$$

where S represents the number of substitutions, D represents the number of deletions, I represents the number of insertions, and C represents the number of correctly recognised words. A lower WER value indicates fewer errors made by the model, thereby indicating better performance.

6.2. Applications of transfer learning and self-supervised learning

In the following section, we have categorised the work of TL and SSL into 11 broad categories. These categories span a wide range of domains, including medical, industrial, satellite, language processing, motion, environment, Web and social networks, speaker, vehicle, robot, and physics. Within each category, the works are subcategorised based on their specific tasks and the type of data used in their training. These subcategories are presented chronologically according to their publication year, providing a structured overview of the subject matter, as demonstrated in Table 2. Within these categories, we provide a comprehensive analysis and summary of various aspects, including the background and context of the investigation, the methodology employed, the intricacies of the training process, and a detailed examination of the experimental results.

6.2.1. Applications in medical field

Medical data analysis is vital for early-stage disease recognition and diagnosis, which can significantly benefit prevention and treatment. However, there are two significant barriers to applying ML methods in this field: concerns about hospital privacy and security and difficulty collecting useful medical data. Due to patient confidentiality, hospitals cannot share personal details, making it difficult to obtain sufficient data to train a robust model from scratch. Public medical image datasets also suffer from the high cost and time required for professional medical annotation. Furthermore, data collection is challenging for rare diseases. As a result, there is a growing demand for developing less data-dependent, more generalisable, and efficient medical diagnoses through ML-based medical image analysis (see Fig. 17).

Xue et al. (2020) conducted a study to examine the performance of mixed TL and Ensemble Learning (EL) techniques in the field of image analysis of cervical histopathology. In pursuit of the decision fusion model, they introduced an innovative framework called Ensembled Transfer Learning (ETL). This framework intelligently retrained three distinct TL architectures, Xception, Inception-V3, VGG-16, and Resnet-50, on the same target dataset, amalgamating them using a weighted voting system to provide optimal results for users. The model's performance was rigorously evaluated on a challenging VEGF dataset, yielding notably positive outcomes across all evaluation metrics compared to every single-member TL pre-training model. This research underscores the notion that combining TL with other advanced training methods not only enhances model performance but also strengthens its stability and generalisability, allowing it to leverage various techniques and handle even the most demanding scenarios effectively. Nevertheless, it is important to note that such a combination method would

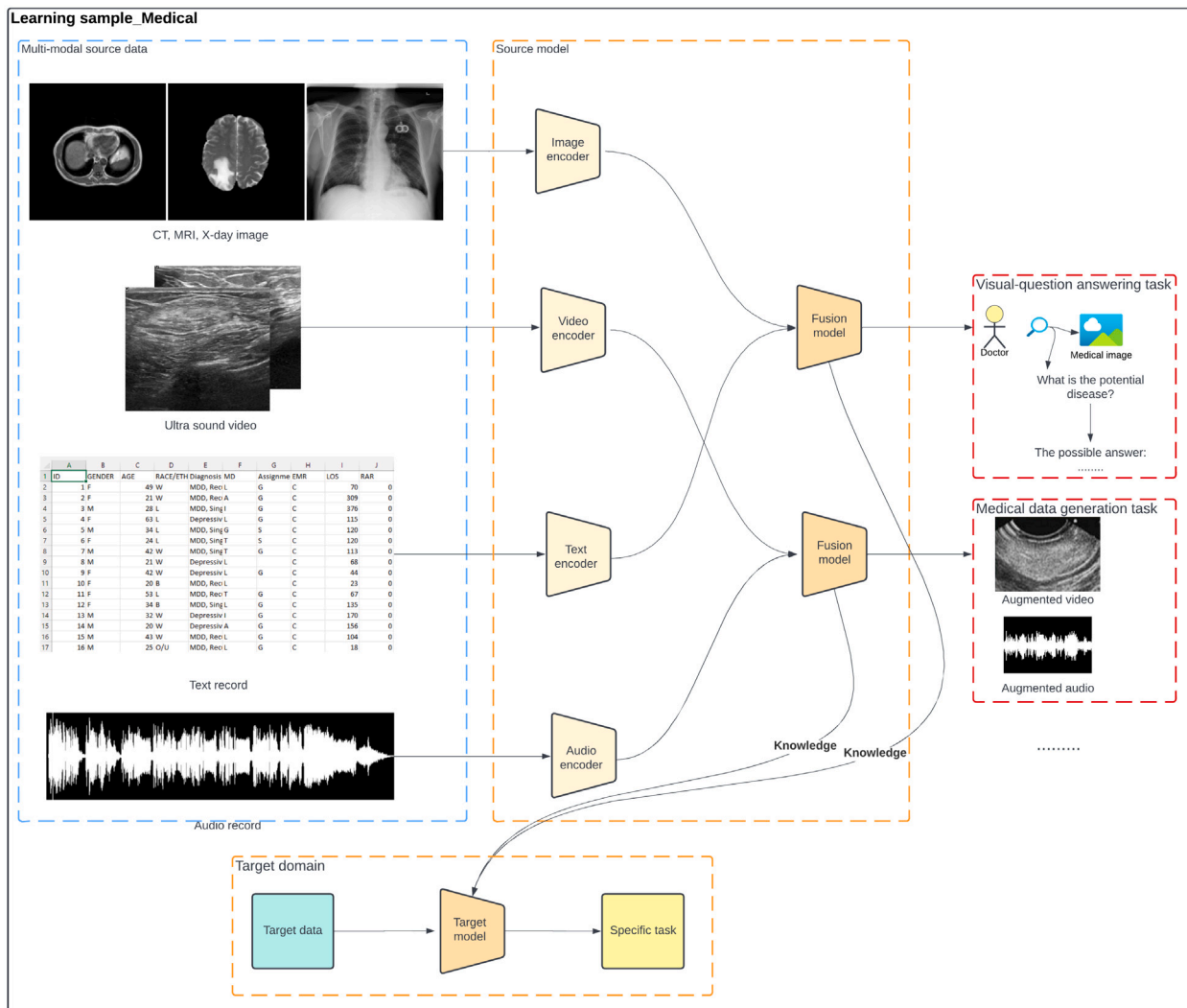


Fig. 17. Multi-modal medical data learning sample.

significantly increase computational resource requirements and impact training speed. This factor is crucially important and requires careful consideration and a balanced approach when designing models.

Lopes et al. (2021) employed the TL method to initially construct a CNN model for sex identification using a substantial source dataset comprising 256,278 patients. Subsequently, they extended the use of this model to address the identification of rare diseases within a relatively smaller PLN dataset, aiding in the identification of PLN patients. A comparison was made between the TL model and two groups of models trained entirely from scratch. The results concluded that pre-training the model for sex classification significantly improved the performance of PLN identification. This not only affirmed the effectiveness of using TL for the detection of rare diseases but also underscored its efficiency in transferring knowledge from broader disease categories to tackle specific and rare conditions.

De Bois et al. (2021) proposed a multi-source adversarial TL framework to learn transferable feature representations across different data sources. The framework, inspired by Ganin et al. (2016), integrates an adversarial training method with TL by training a feature extractor module to produce domain-invariant features. Furthermore, a domain discriminator module was employed to distinguish between the source and target domains. From the result, this study demonstrated the effectiveness of their adversarial pre-training approach in improving the transfer quality and achieving generalisable feature representations

that can be transferred to unseen realms. This study has proved the efficiency of the TL method in extracting valuable and meaningful medical features.

Fedorov et al. (2021) highlighted the drawback of traditional supervised training, which is susceptible to learning from irrelevant features, thus diminishing the relevance of the acquired features in subsequent analysis. In response to this limitation, the researchers used contrastive SSL techniques to train a model using fMRI and magnetic resonance images for the classification of Alzheimer's disease. They also examined the effectiveness of multimodal architectures and data fusion methods in enhancing downstream classification performance for both modalities through their experiments. This study has illuminated the advantages of the SSL method in training with multimodal data. Specifically, it provides valuable regularisation through modality interactions, resulting in not only improved model performance but also significantly enhanced generalisability. Furthermore, the SSL method exhibits substantial potential in feature extraction and data fusion.

In their work, Bargshady et al. (2022) proposed a novel approach for COVID-19 detection that combines a TL model based on Inception V3 with a CycleGAN-based data augmentation method. The TL technique was used to leverage the knowledge learnt from a COVID-19 chest image and X-ray CT dataset and enhance the model's performance with a limited amount of labelled data. By applying their approach, the researchers achieved a high accuracy of 94.2% for COVID-19 detection.

This outcome underscores the distinct advantage of the TL method, which leverages pre-existing knowledge to expedite the training of new models and provide rapid responses to emerging diseases.

Gardner et al. (2022) directed their efforts towards addressing a critical issue in the development of structural health monitoring systems: the scarcity of available health-state data. To overcome this challenge, they utilised kernelized Bayesian TL, a heterogeneous transfer learning method that aims to leverage information across multiple datasets, to generalise for a population of structures and transfer label information from domains with similar designs. They trained a discriminative classifier by mapping inconsistent and heterogeneous feature data in a shared latent space. This is the first application of a heterogeneous transfer learning method in the field of structural health monitoring, and experimental results unequivocally demonstrate the feasibility of sharing and transferring label information within structural data, thus showcasing the good potential of the TL method in the domain of structural health monitoring.

Kathamuthu et al. (2023) conducted a study aimed at improving COVID-19 detection using CT scan images through CNN frameworks enhanced by conceptual transfer learning. They examined and evaluated various CNN approaches based on TL, including VGG16, VGG19, Densenet121, InceptionV3, Xception, and Resnet50. The researchers ensured consistency in training batch size, epochs, learning rate, and other hyperparameters to facilitate a fair comparison among these different CNN models. The experimental results highlighted the exceptional performance of TL-based VGG16, achieving an impressive accuracy of 98%, surpassing both the other models tested and state-of-the-art approaches. This investigation further demonstrated the effectiveness and advantages of TL methods in handling newly emerged infectious diseases.

Jiao et al. (2020) marked the pioneering exploration of learning foetal ultrasound video representation through the use of the SSL method. The model was tasked with solving anatomy-aware challenges, including reordering shuffled video clips and predicting geometric transformations, all without explicit supervision. The findings of this investigation notably underscored the efficacy of SSL in acquiring meaningful and robust representations from unlabelled data. Furthermore, it demonstrated the seamless transferability of these extracted features to downstream tasks. This outcome underscores a noteworthy advantage of the SSL method, prompting consideration for its testing and application across various other medical modalities, thus offering a promising avenue for future research.

In their study, Xue and Salim (2021) investigated a novel SSL approach for the detection of COVID-19. Instead of utilising a conventional image dataset for classification, the researchers used unlabelled audio cough data to train a feature encoder based on transformers. They also developed a random masking technique to learn resilient representations of respiratory sounds. Furthermore, the study explored an ensemble framework for training downstream tasks, employing model fusion techniques with varying model structures such as VGG and GRU. Extensive evaluations conclusively demonstrated that their proposed ensemble network, coupled with random masking, achieved top-tier performance. This research underscores the advantages of employing the SSL method in medical applications. It enables deep learning models to harness previously challenging data types, particularly those that are difficult to annotate manually. Moreover, it highlights the immense potential of the SSL method for enhancing performance by synergising with other advanced deep-learning techniques.

McCreery et al. (2020) proposed a double fine-tuning TL approach for a COVID-19 interactive online FAQ system. Their approach involved pre-training a neural network training on medical question-answer pairs, followed by fine-tuning on medical question-question pairs task to determine medical question similarity. The whole system was then double fine-tuned again for the ultimate goal of maximising the medical question similarity. The model's performance was evaluated in terms of

accuracy, and it achieved a 5% improvement compared to other pre-training tasks in matching user questions to related COVID questions, which demonstrated the particular advantages of the double fine-tuning SSL method towards complex semantic tasks. As a current limitation and future research direction, the authors mentioned it is necessary to use weak supervision, augmentation rules, and even active learning to enforce the number of training examples in those difficult regions of the data to improve the model's efficiency.

Apart from commonly used image and language data, tabular data's feature correlation structure is often unknown and varies across different datasets, making it challenging in information extraction. To address this issue, Yoon's research team (Yoon et al., 2020) proposed a novel tabular data augmentation method an SSL approach to learn vector information from corrupted tabular data. Specifically, the SSL model was pre-trained with two clinical tabular datasets to solve the pretext predictive tasks with the masked medical tabular records. The SSL model used two encoders called mask vector estimator and feature mask estimator, aiming to learn vector information and feature embeddings. The experimental results demonstrate that their framework surpasses the performance of previous supervised and semi-supervised learning models. This success underscores the potential effectiveness of SSL methods when applied to tabular data, which can serve as a source of inspiration for future research in this domain.

Ankenbrand et al. (2021) aims to investigate the impact of both data type and quantity when constructing an effective knowledge transfer model. To assess this, they initially trained a neural network on a publicly available imaging dataset to focus on challenging segmentation tasks involving the left ventricle and myocardium in cardiac cine MRI. To augment their dataset, they also employed specialised, publicly available models to generate labels for the target dataset, to gain more access to the annotated data. They then analysed the TL method using the trained network as a starting point to segment the 7T cine data. The experiments showed that introducing the TL method helped reduce the required training data by 90% without negatively affecting the segmentation performance. This outcome underscores a significant advantage of the TL method in diminishing the data prerequisites for training valuable deep learning models. However, the study's report also raised a noteworthy concern. It pointed out that the specialised model design had somewhat diminished the model's capacity for generalisation. Striking a balance between optimising single-application performance and preserving the generalisability of the pre-trained model remains a prominent challenge.

Zhao et al. (2021) proposed a TL-based algorithm that utilised U-Net for fundus image segmentation. Inspired by the previous work in combining high-resolution local features with low-resolution global features, they proposed a U-Net architecture combining TL methods with attention. To test the proposed model, they trained the network on DRIONS-DB and Drishti-GS datasets to obtain encoder weights. Subsequently, they utilised the trained weights to perform fundus image segmentation tasks. As a result, the trained attention U-Net model can be effectively used for a fundus image dataset with only limited labels while still ensuring high segmentation accuracy. Furthermore, they observed that the model can significantly reduce the model inference time, with a reduction of more than 50% of time consumption.

Azizi et al. (2023) conducted a study with the objective of utilising out-of-distribution data from distant domains in downstream medical imaging tasks. To address this complex task, they proposed a strategy called RENEDIS. The approach involves two main steps: first, adopting large-scale supervised TL on natural images, and second, transferring the acquired knowledge to the intermediate domain using contrastive SSL on medical images. In the final stage, the pre-trained model can easily achieve robust performance while minimising the data requirements for task-specific training. Experiments were conducted across six medical domains, showcasing the superiority of the proposed SSL model over strong supervised baseline models. In particular, their approach only requires a small amount of data to achieve performance

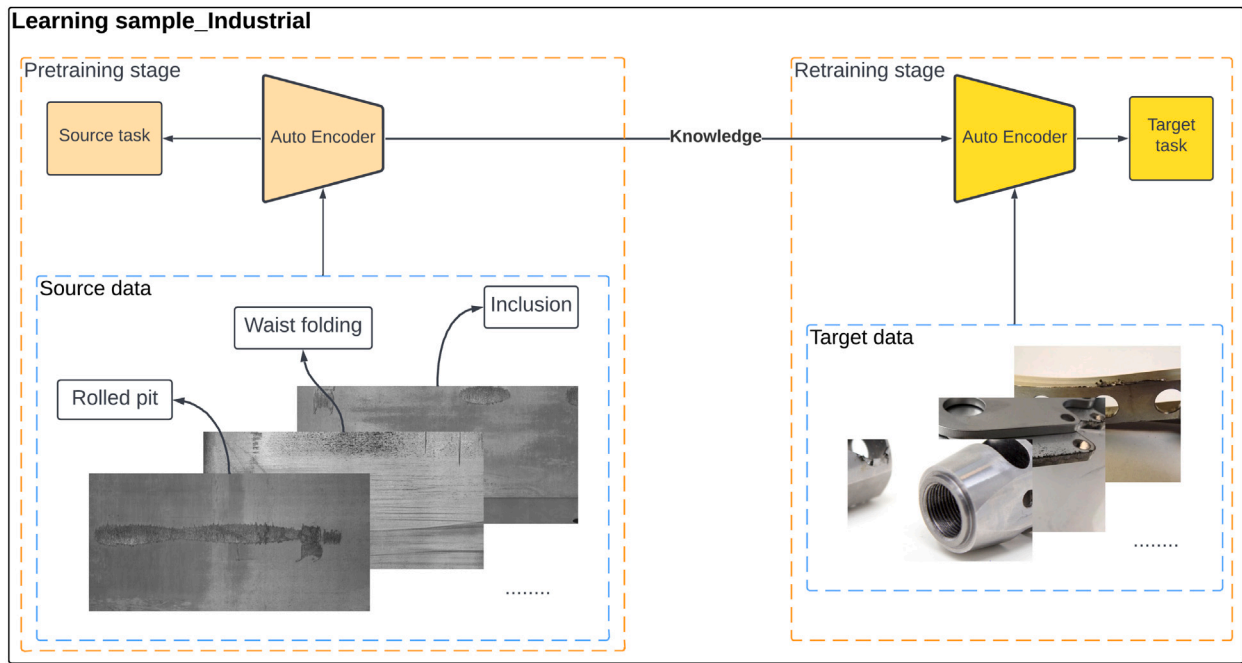


Fig. 18. Industrial equipment diagnosis sample.

comparable to that of fully supervised models, showing great potential in real-world applications.

Moor et al. (2023) conducted a pioneering study in the field of medical AI, introducing the concept of Generalist Medical AI (GMAI). GMAI aims to develop a reusable self-supervised model capable of addressing a wide range of tasks with minimal or no task-specific labelled data. To achieve this, the researchers leveraged large-scale and diverse multi-modal datasets, encompassing various types of data such as images, electronic records, and medical text. By employing SSL, the model learned rich representations, reducing the need for manual labelling efforts. Despite the promising potential of GMAI, the research is still in its early stages and encounters several challenges. These challenges include the demand for massive amounts of data, substantial computational costs, and the inherent complexity of creating a versatile model that can be effectively trained and comprehended. Nevertheless, the medical field would greatly benefit from the development of a true generalist model that can be applied across numerous tasks without the need for frequent retraining from scratch.

Based on the paper we reviewed so far, TL models have generally shown advantages in improving the model's performance (Bargshady et al., 2022), extracting meaningful and transferable features (De Bois et al., 2021), accelerating training and prediction speed (Xue et al., 2020; Zhao et al., 2021), and reducing the data requirement and risk of overfitting in most of the scenarios (Lopes et al., 2021). Furthermore, the TL-based models have demonstrated a promising capability in solving various medical tasks, including classification, segmentation, and generation (Gardner et al., 2022; Kathamuthu et al., 2023; McCreery et al., 2020). However, several limitations need to be addressed. One of these limitations is the high computational cost and workload associated with the multi-stage training required to transfer knowledge in TL methods (Xue et al., 2020). Furthermore, Ankenbrand et al. (2021) have expressed concern that TL approaches need to transfer related knowledge to achieve the promised performance. For instance, using models pre-trained on the ImageNet dataset may not provide satisfactory performance for downstream tasks in the medical field, as the general knowledge may not be relevant to solving specific tasks.

At the same time, we observed a smaller volume of research in the field of SSL for medical applications compared to TL, partly due to the lack of domain-specific designed pretext tasks and the past success

of the TL method in this research direction. Nevertheless, current SSL studies have already illuminated numerous advantages. For instance, SSL methods exhibit robust performance even when confronted with a limited amount of labelled data, effectively addressing the challenges of few-shot learning that often pose difficulties for supervised models (Azizi et al., 2023). Furthermore, the ability of SSL to learn from unlabelled data, without relying on external labels, proves especially beneficial in medical fields where obtaining labelled data can be scarce or logistically challenging (Xue & Salim, 2021). Moreover, SSL methods have shown further advantages when working with multimodal data and enhancing a model's generalisation capability (Fedorov et al., 2021; Jiao et al., 2020). Yoon et al. (2020) also proved the potential of the SSL method in handling structure data. It is important to acknowledge that one limitation associated with SSL is the relatively high time consumption and computational cost, stemming from the nature of SSL's reliance on pseudo labels for training. So far, it is hard to determine if the SSL methods have established a definitive advantage over TL methods in medical applications, necessitating further research for a more comprehensive understanding of their comparative strengths and limitations.

6.2.2. Applications in industrial field

In the industrial domain, equipment degradation diagnosis traditionally relies on experienced engineers and cannot be carried out frequently due to the need to shut down machines for worker safety. ML techniques have been explored as a potential replacement for manual recognition, but acquiring enough labelled training data remains a significant challenge. The difficulty arises from manufacturers holding different patents, resulting in equipment size and appearance variations. This inconsistency increases the difficulty of recognising the equipment and requires additional costs for manual annotations (see Fig. 18).

Zhang et al. (2020) developed an ensemble-based deep TL network to address the challenge of recognising degradation by inadequately labelled training data. Their approach involves utilising a set of selected instances from a source domain to train multiple stacked autoencoders with different activation functions. The extracted features from these encoders are then inputted into a support vector machine to construct a complete model using an ensemble approach. The experimental results

unequivocally show that the use of the TL-based approach surpasses the performance of other conventional methods. Furthermore, the transferred model significantly improves accuracy in the target domain, particularly when there is an insufficiency of labelled target data.

Using acoustic emission signals, [Xin et al. \(2020\)](#) developed a TL method to identify fractures within stay-cables. They first trained a CNN model to identify acoustic emission-monitored signals. Then they used the trained network to bridge the relationship between the acoustic emission signals and the cable status. The monitoring of acoustic emission signals essentially boiled down to a binary classification task. Through experimental evaluation, this study demonstrated the effectiveness of transferring auditory emission signal information for fracture identification. Furthermore, the proposed TL method successfully mitigated the challenge of data scarcity commonly encountered in the acoustic emission monitoring field.

[Zhu et al. \(2021\)](#) addressed the issue of chiller fault detection and diagnosis, where most prior works concentrated on training fault diagnostic models for specific chillers instead of developing a generic framework for detecting defects in different types of chillers. To overcome this limitation, they proposed a heterogeneous transfer learning approach using a domain adversarial neural network to generate the diagnostic model. The model was first trained on an information-rich chiller with normal and fault data and then retrained on the target chiller with different features. The performance of this approach was compared with five conventional ML models and demonstrated significant advantages in terms of performance and potential for practical application.

[Michau and Fink \(2021\)](#) identified a challenge in industrial anomaly detection lies in the demand for ample and representative data for training, while concurrently grappling with rapidly changing operating conditions within the system. Compounding this issue is the protracted time required for data collection. To bridge this gap, researchers investigated the use of unsupervised transfer learning for the detection of industrial anomalies. Their proposed method entails a multi-component architecture, comprising a feature encoder, an adversarial feature domain discriminator, and a feature processor. These components work cohesively to extract meaningful features from unlabelled data. The feature extractor underwent supervised training to ensure its ability to capture meaningful features, while the discriminator played a vital role in minimising multidimensional scaling loss, thereby facilitating the transfer of label knowledge from the source domain to the target domain features. This approach enables the model to effectively handle unlabelled data, thus enhancing anomaly detection in industrial settings. However, the study identified a limitation in which the anomaly detector could not recognise the primary class when there is a domain change in the various data sources, such as different engine setups or different nature images.

[Asanuma et al. \(2020\)](#) sought to improve the effectiveness of genetic algorithms in the development of high-performance electric motors, particularly in the context of topology optimisation under complex real-world conditions. To address the challenge of limited data availability, they proposed a TL model that uses transfer tuning for the CNN model to expedite the evaluation process. To be specific, their approach involved initially training the network with generic image data and subsequently fine-tuning it using various electric motor images to tailor it to the specific task. The experimental results highlight a compelling advantage of the TL method: it significantly reduces computational costs by approximately 13% to 15% when compared to traditional conventional methods. This underscores the TL method's capacity to conserve valuable computational resources.

[Ren et al. \(2021\)](#) introduced a self-supervised long-short-term memory (LSTM) deep factorisation machine model to solve the challenge of analysing complex industrial data with diverse characteristics, nonlinear relationships and massive unlabelled samples. The proposed SSL model utilised an unsupervised LSTM model with two self-supervised mask strategies to capture the interdependence between features and

dynamic fluctuations in time series. The results indicate that the proposed method outperforms the stacked autoencoder-based models and achieves state-of-the-art performance in this area.

[Akrim et al. \(2023\)](#) tackled the challenge of fatigue damage prognostics by employing a few-shot SSL approach. In their study, the researchers used a substantial amount of unlabelled training data to perform data augmentation and implement self-supervision techniques. This strategy aimed to overcome the limitations posed by a scarcity of labelled data on industrial systems. The learned parameters from the self-supervised pre-training phase were then transferred to the downstream task learning stage, where the model was fine-tuned using only a small number of labelled samples from the target domain. The experimental results demonstrated the superior performance of the self-supervised pre-trained model compared to non-pretrained models. Additionally, this approach exhibited the potential to reduce both the computational costs and the labelled data requirements during downstream task learning.

[Zhang et al. \(2023\)](#) identified a critical challenge in machinery fault diagnosis, specifically the insufficiency of high-quality and sufficient data from a single industrial user. To overcome this limitation, the researchers introduced a novel approach called blockchain-based decentralised federated transfer learning. This method enables the collaborative construction of models by multiple clients. The system consists of two key components: the multiple clients and the blockchain. In the proposed framework, the federated learning task for fault diagnosis is distributed to all participating clients, each with an initialised model. The optimisation process then commences through iterative iterations. Importantly, this approach ensures the protection of local private data, as the trained parameters are not directly shared with the source data during model communication. The utilisation of blockchain technology enhances the security and privacy of the federated transfer learning process in machinery fault diagnosis.

Traditional ML methods have often proven ineffective in industrial settings due to the recurring problem of data scarcity and the complexities involved in processing comprehensive datasets. The adoption of the TL techniques has substantially alleviated this issue ([Xin et al., 2020](#)), as evidenced by experimental results showcasing notable improvements in performance ([Zhang et al., 2020](#); [Zhu et al., 2021](#)), and reductions in computational costs ([Asanuma et al., 2020](#)). Additionally, heterogeneous transfer learning methods have been extensively used in the industrial field ([Zhu et al., 2021](#)) since they allow for transferring knowledge from source domains presented in different feature spaces from the target domain. However, there is also a limitation reported in TL research which is the presence of unrelated or irrelevant source instances in real-world applications ([Michau & Fink, 2021](#)), making a direct transfer from source data to the target data less suitable.

The amount of observed research in the industrial field remains relatively limited when compared to the TL studies. However, recent studies have demonstrated the potential of SSL in challenging tasks such as predicting industrial data characteristics ([Ren et al., 2021](#)), and emphasised the advantage of the SSL method in acquiring meaningful features without a heavy reliance on copious manual annotations ([Akrim et al., 2023](#)). Nevertheless, the SSL method must overcome the obstacle of non-standardisation in the industrial field and train models to learn more generalised features that can be applied to new machines' predictions.

6.2.3. Applications in remote sensing field

In the field of remote sensing, data is typically acquired in the form of satellite images. High-quality satellite images play a crucial role in various applications, such as weather prediction, river detection, and terrain observation. These images are captured by sensors onboard satellites and can exhibit varying colouration or greyscale appearance, depending on the type of sensors used. Furthermore, these images are characterised by high resolution, well-organised structures, timeliness, and the presence of multiple objects that require annotation.

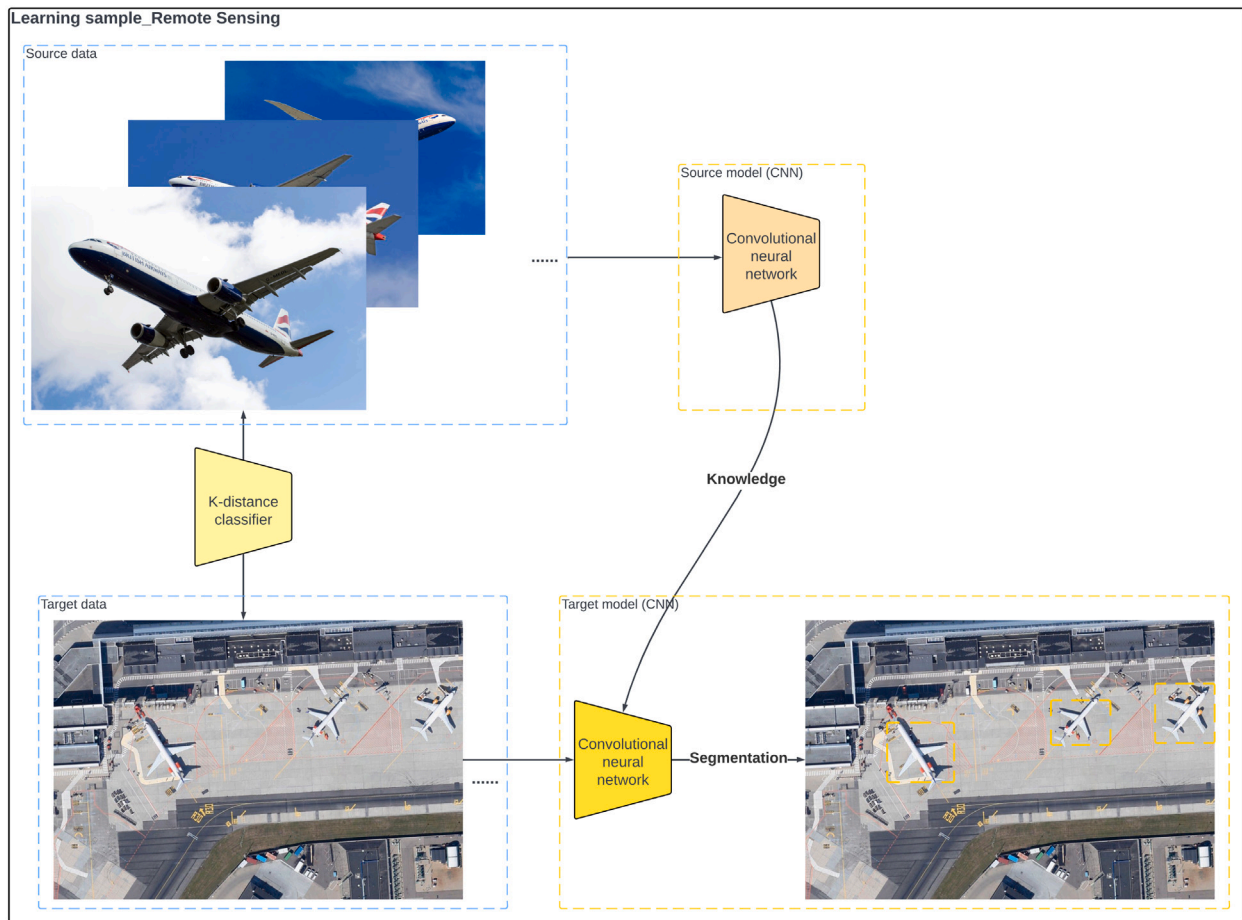


Fig. 19. Aeroplane segmentation sample.

Recently, there has been a surge in research efforts aimed at integrating artificial intelligence technology into remote sensing, with a primary focus on tasks related to satellite imagery analysis. However, the significant cost associated with acquiring large-scale labelled satellite datasets has posed a substantial barrier to researchers seeking to enhance model performance. Addressing this challenge has become an urgent priority within the field (see Fig. 19).

Yuan and Lin (2020) have directed their attention towards the classification of satellite image time series, a pivotal area of research in remote sensing with a wide range of applications. The authors have introduced an SSL-based approach utilising extensive unlabelled data for forecasting time series data. Their approach is based on a transformer model, designed to extract informative and distinguishing features from unlabelled satellite data, while also learning the underlying temporal patterns and land cover semantics. It is worth noting that this study has primarily focused on the spectral profiles of individual pixels, overlooking spatial correlation information during the model's pre-training phase. Nevertheless, the experimental results have convincingly demonstrated that the transfer of spectral-temporal features from unlabelled satellite data can have a positive impact on subsequent task learning. This approach has also shown the potential to enhance the generalisation capabilities of deep learning models, particularly for specific tasks related to the classification of time series satellite images.

Li et al. (2023) have undertaken an exploration of the few-shot remote sensing scene classification task. Their research addresses the challenge that most self-supervised pretext tasks face, which is the problem of high interclass similarity. Additionally, searching for the

most effective pretext task among numerous possibilities can be a time-consuming and computationally expensive process. To tackle these issues, the authors have introduced an innovative ensemble SSL framework, which is built upon a TL-based few-shot scheme and incorporates a novel global-local contrastive learning auxiliary task to address the issue of low interclass separability. The core of their training approach involves simultaneously training multiple backbone TL-based networks using various random combinations of self-supervised auxiliary tasks. The features extracted from these networks are then combined to enhance the generalisation of the model. Experimental results have convincingly shown that the multi-form SSL framework achieves state-of-the-art performance in few-shot remote sensing scene classification. Importantly, it achieves this with significantly lower time complexity, reducing time costs, and outperforming exhaustive methods in few-shot classification performance. This study not only introduces a specifically designed pretext task and a versatile ensemble enhancement method but also underscores the importance of pretext tasks in SSL. These findings have the potential to influence and improve various SSL methods.

Muhtar et al. (2022) have undertaken a study focusing on the semantic segmentation of remote sensing images. This is a challenging task because obtaining pixel-level annotations for such images can be extremely difficult. To address this challenge, the authors have employed the SSL method to acquire valuable representations. However, they argue that traditional SSL approaches pre-trained on ImageNet data are not suitable for this specific task. This is because these approaches are not designed to capture both spatial position and semantic information from images. Drawing inspiration from semantic segmentation ground truth masks, the authors have introduced a pretext task

that involves indexing the pixel positions and generating an index mask for each view, prompting the model to learn to make predictions. Additionally, they have used remote sensing-related data to train their SSL model. The final results were evaluated and compared with other SSL methods based on ImageNet, while these results have showcased state-of-the-art performance in the semantic segmentation task, highlighting the importance of well-designed pretext tasks in extracting valuable features.

Chen et al. (2022) have highlighted a significant challenge in the realm of remote sensing images, which is the small volume of data. This abundance of data has posed difficulties for deep learning methods, often leading to issues like overfitting and detection problems. To tackle this challenge, the authors have adopted a TL approach. They leverage TL to acquire features and structures from natural images, which are then transferred to the task of detecting aeroplanes in the remote sensing domain. Recognising the distinctions between source natural images and the target satellite images, the authors have also employed a K-distance method to assess the similarity of images and instances across these domains. Through a series of adaptation experiments that involve training on both high-resolution and lower quality satellite images, their TL-based R-CNN model has demonstrated its effectiveness in terms of performance and generalisation.

Gomroki et al. (2023) have focused their research on change detection in remote sensing data, particularly in areas where detecting changes is crucial. Traditional supervised deep-learning approaches have been time-consuming due to the difficulty of collecting annotated data. On the other hand, unsupervised learning methods that rely on pixel-level learning have been sensitive to radiometric correction challenges. To tackle these issues, the authors have introduced a semi-TL approach. The model employed an ensemble deep learning model that combines a supervised pre-trained EfficientNetV2 model as the encoder and an unsupervised U-net model as the decoder. This combination allows the model to efficiently learn distinctive features and detect pixel-level changes in satellite images. The experimental results not only showed an improvement in training speed but also demonstrated enhanced model performance in terms of accuracy and F1 scores. This work proved the TL method as a valuable foundation for future advancements.

Xiao et al. (2023) have delved into a popular area of remote sensing, which is single image super-resolution. This task has long grappled with challenges related to the availability of real-world datasets and the development of accurate degradation estimation methods. To address these issues, the authors have introduced an innovative approach known as SSL degradation-guided adaptive network. This method is designed to bridge the gap between simulated and real-world domains. Specifically, it leverages self-supervised contrastive learning techniques to acquire prior information. It achieves this by generating pseudo images to identify similar degradation patches, learn related degraded features, and construct embeddings. In their experiments, the researchers applied this SSL model to superresolve ground-truth images and predict their latent high-resolution counterparts. The results demonstrated that the SSL model effectively adapted its features to a test target domain, allowing precise degradation estimation without the need for manual labels. This work represents a significant advancement of the TL method in the problem of data scarcity in single image super-resolution within the field of remote sensing.

In the realm of deep learning-related remote sensing studies, several recurring limitations have been extensively discussed. These limitations include the persistent challenge of data scarcity in satellite images, the difficulty associated with annotating these images, and the absence of tailored methods for effectively extracting valuable information. Remote sensing tasks demand a unique knowledge base compared to other fields due to the wealth of information contained within a single image. This complexity often poses challenges for applying pre-training methods in remote sensing. Recent endeavors in the remote sensing domain, as observed in studies such as Muhtar et al. (2022),

have showcased the benefits of TL and SSL methods. These advantages include improvements in model performance, as well as enhanced generalisation capabilities (Chen et al., 2022), and mitigated data scarcity concerns (Yuan & Lin, 2020).

However, it is important to acknowledge that the limitations arise from the distinctive characteristics of satellite images, making the application of traditional pre-trained methods challenging. Differences in data distribution and features between satellite images and commonly used datasets like ImageNet can lead to subpar performance and potential domain mismatches that degrade model performance. Consequently, recent research efforts have concentrated on employing ensemble methods to address these limitations. This includes techniques such as combining domain adaptation strategies to maximise domain similarity (Xiao et al., 2023) and using multiple pre-trained methods to strike a balance between performance and training costs (Gomroki et al., 2023; Li et al., 2023). These approaches not only offer potential solutions to improve pre-trained methods within the remote sensing domain but also hold promise to improve their applicability in diverse fields of study.

6.2.4. Applications in natural language processing field

Natural language processing (NLP) encompasses a broad range of applications, including but not limited to question-answering, text generation, information retrieval and text detection. The general objective of NLP is to enable computers to understand and process human language. While SSL was not as widely adopted in computer vision research, it has made significant progress in NLP. BERT (Devlin, Chang, Lee, & Toutanova, 2018), a popular NLP algorithm, is a prime example of a successful self-supervised model. However, in the NLP field, it is hard to obtain fully annotated datasets due to the large size of the text data and the resources required for annotation. This lack of high-quality annotations and the challenge of improving generalisation properties has been a long-standing limitation in the field (see Fig. 20).

Liu et al. (2020) identified a possible issue with the majority of current state-of-the-art chatbots, as they used a turn-by-turn interaction approach. This means the chatbots respond to each customer's utterance without considering its meaning. To tackle this problem, they put forward an SSL-based multi-turn response triggering model. The model is trained on extensive human-to-human dialogues to develop semantic matching and determine whether the given context warrants a response from the chatbot. Even when the model is not explicitly tailored for performance enhancement, the findings demonstrate that multi-turn response triggering methods exert a discernible influence. Factors such as data scale, sliding window size, and the incorporation of weight loss accumulated attention have been shown to have a substantial impact, resulting in significant improvements in model performance.

Cao et al. (2020) introduced a self-supervised domain adaptive framework to enhance the model's domain adaptability in automated essay scoring applications. They noted that existing state-of-the-art essay scoring models neglected domain discrepancies and struggled to utilise data from diverse domains. To address this, the authors devised two innovative self-supervised pretext tasks, sentence reordering and noise identification, to extract valuable information that could be applied across different domains. This study aims to further improve the generalisability of SSL models and to avoid domain mismatch. The domain adversarial training technique implemented in this study has been tested and shown effectiveness in improving the robustness of the model to domain variation.

Van Nguyen et al. (2021) introduced an innovative approach to cross-lingual alignment for relation and event extraction tasks. To link the source and target languages, the model computes representation vectors for classes and word categories through contextualised representations. Additionally, the authors suggested learning two versions of representation vectors for each class and aligning them to enable class-specific alignment of cross-lingual representations. Their method

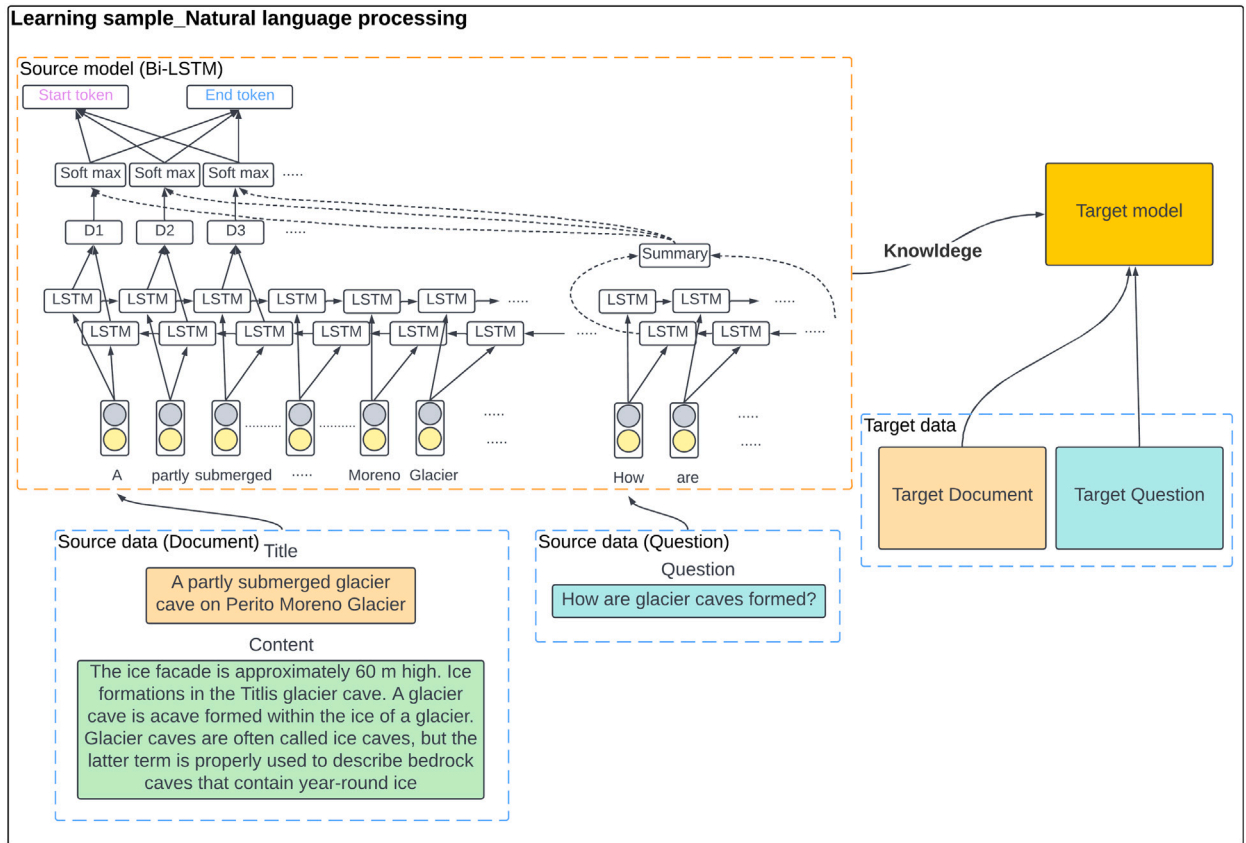


Fig. 20. Question answering system sample.

exhibited advantageous outcomes through cross-lingual experiments involving English, Chinese, and Arabic for REE tasks.

In Omran's research (Omran et al., 2023), the challenges encountered in Arabic language machine learning were discussed, including limited data availability, the need for multi-lingual deep learning models, and the lack of specific studies on Arabic dialects. To overcome these challenges, the researchers utilised TL techniques to transfer knowledge from large-scale English datasets to help build an Arabic language model, using the LSTM model as the basis. Through experiments conducted on various datasets encompassing different Arabic dialects, the results showcased the advantage performance of TL models in transferring knowledge between English and Arabic languages.

Lu et al. (2021) developed a sequence-to-structure network for integrated event extraction tasks, while the challenge stemmed from the intricate structure of event records and the semantic divide between textual content and event representations. The research team featured a constrained decoding algorithm for event knowledge integration and a curriculum learning algorithm for model training. The model processes raw text inputs and generates event structures through a transformer-based encoder-decoder architecture. The results of the experiments indicated that the use of a single universal model for modelling tasks and predicting multiple labels helped the model achieve competitive performance in the TL setting.

Elmadany et al. (2022) delved into the realm of multilingual translation tasks, focusing on the language pair of English and Arabic. Their innovative framework utilised a unified transformer framework called T5, known for its successful TL capabilities, and pre-trained this model with an extensive corpus of Arabic text data. Subsequently, they conducted comprehensive evaluations of the pre-trained model using four Arabic text datasets, encompassing tasks related to text translation, summarisation, and generation. The experiments revealed that the use of newly pre-trained tasks considerably improved the performance of

the model. The most notable was the revelation that the proposed TL model achieved a performance comparable to the benchmark state-of-the-art deep learning model while reducing the target data requirement by a remarkable 50%. This underscores the efficiency and efficacy of the TL approach in multilingual translation tasks.

Xu et al. (2021) researched a multi-modality learning task called automated audio captioning, which aims to generate concise text summaries for audio clips. They proposed a topic model for audio descriptions and utilised a TL scheme to extract audio representations and build embeddings. Through experiments, they compared two pre-training tasks and two audio encoder architectures including CNN10 and CRNN5, and found that local information and abstract representation learning are more critical than global information and temporal relationship learning within audio captioning tasks.

In their work, Sung et al. (2022) identified a common limitation of pre-trained language models as the rapid expansion of their pre-trained parameter sets. This growth in parameter size poses a challenge when attempting to transfer such model weights entirely. To address this issue, they presented a newly designed parameter-efficient TL framework that incorporated adapters into the pre-trained model. These adapters are essentially sub-networks characterised by small parameter sets, strategically inserted after each attention and feed-forward layer within the model. Their primary function is to use the downsampling and upsampling layers to adapt the data distribution and reduce the number of parameters that need updates when adapting the model to downstream tasks. The experiment encompassed the evaluation of three adapter-based approaches, particularly in the context of image-text and video-text tasks, and demonstrated the effectiveness of weight-sharing techniques, which improved efficiency and performance by enabling the sharing of weights and knowledge across tasks.

In the NLP field, a notable distinction exists when compared to domains like medicine and industry that are renowned for data scarcity.

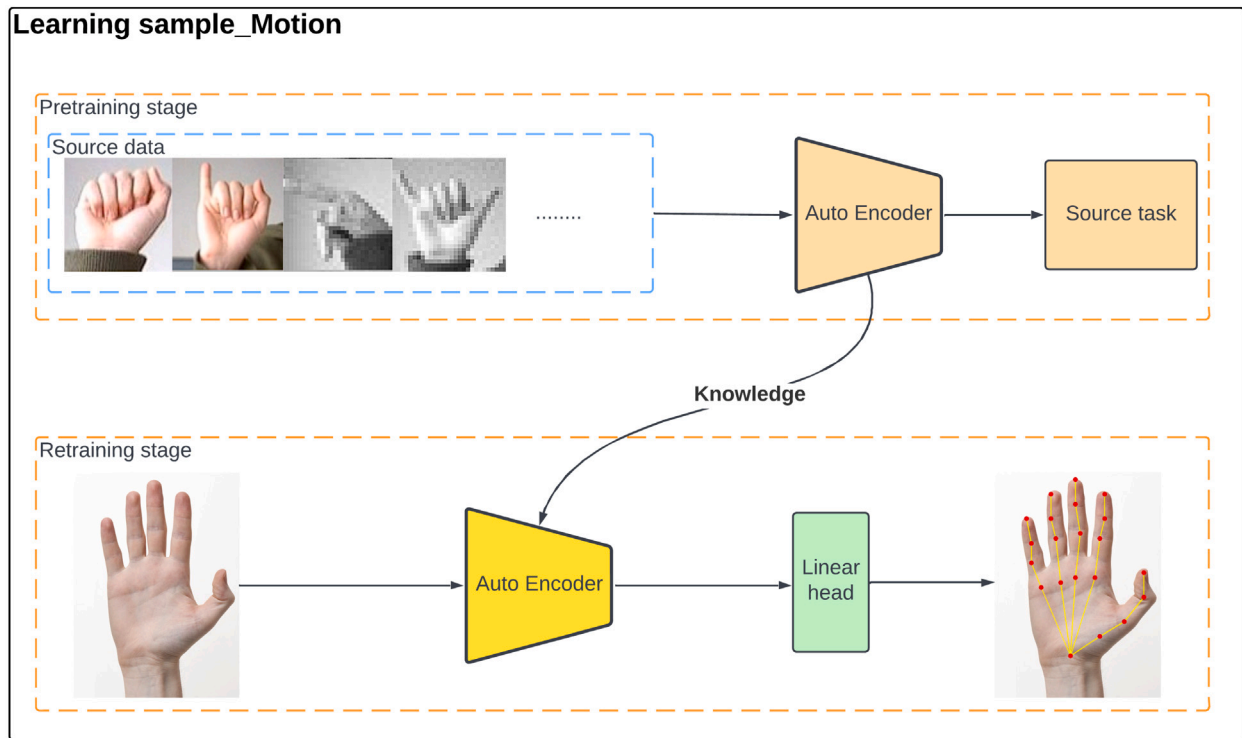


Fig. 21. Hand pose estimation sample.

While acquiring common text data is relatively straightforward in NLP, the principal challenge lies in comprehending the intricate semantic relationships and sentence structures. As highlighted in this review, TL methods have become increasingly prevalent in conjunction with unsupervised language models such as LSTM, BERT, and Transformer. Lu et al. (2021) and Omran et al. (2023). The utilising of the hybrid model also shows the potential of the TL approach to learning from multi-modality data (Xu et al., 2021). The key benefit of the TL method is its ability to generate embeddings and transfer related syntactic and semantic knowledge to build a bridge between different languages and enhance the performance of the target tasks (Elmadany et al., 2022; Van Nguyen et al., 2021). However, it is essential to acknowledge certain limitations. Training multistage TL models can be computationally intensive and memory-inefficient, as observed in the study (Sung et al., 2022). Furthermore, some researchers prioritise the collection of large training datasets, which may inadvertently slow knowledge transfer and pose challenges in terms of data storage and processing speed.

In the same situation, the size of the training datasets used in SSL has become sufficiently large to even reach a bottleneck. As such, simply increasing the size of datasets is unlikely to yield significant performance gains. Liu et al. (2020) noted that when the training dataset reached a scale of 200k, the model performance stabilised. Based on recent research (Cao et al., 2020), SSL methods have shown great potential in this field, as the unsupervised structure of SSL makes it more suitable for handling unlabelled text data. Another notable observation from both previous industrial applications and this NLP application is the utilisation of adversarial training to mitigate the discriminative loss between the source and target domains. This approach effectively addresses the challenge of domain variation and improves the robustness of the model (Cao et al., 2020; Zhu et al., 2021). The positive results in these studies suggest that adversarial training represents a promising research direction to overcome the common limitations associated with pre-training methods. To move forward, efforts should focus on improving the quality of data and the generalisability of the model.

6.2.5. Applications in motion field

In motion field, information can manifest in various media, such as the pixel movement in videos or the actions depicted in 3D scan images. Nevertheless, annotating such data can be challenging due to the requirement of segmenting subjects and identifying various actions at the same time. Moreover, annotating certain recording media, such as videos, can be extremely time-consuming. To mitigate the high cost of human labour required for annotating, the field has turned to pre-trained methods, with a particular emphasis on the SSL method. These methods harness self-generated labels, significantly alleviating the need for extensive manual annotation efforts (see Fig. 21).

Li et al. (2020) introduced a new SSL technique for 3D human pose estimation from monocular images, since they trained the network solely based on geometric knowledge without additional manual annotations. Their approach consists of a two-stage pipeline that converts 2D images into 3D poses and trains the network to learn the scale information and support precise 3D pose predictions. The researchers conducted evaluations by comparing their method against recent state-of-the-art weak and self-supervised learning methods within the H36M and 3DHP datasets. The experimental results demonstrated the superiority of their proposed method, as it established a new state-of-the-art in this particular application.

Bhatnagar et al. (2020) highlighted the unreliability of traditional supervised human models, which can only be initialised when they are close to the solution and lack end-to-end differentiability. To address this problem, the authors devised a human pose detection model based on a self-supervised loop. The loop consists of a neural network that predicts the position of each scan point on the human model's surface, followed by a parameterisation of the model to transform the corresponding points back to scan-based parameters, which constitutes a round. Their results show that the proposed model improves accuracy, as it allows the processing of additional unlabelled raw scans within the regression loop.

In their work, Dai et al. (2020) proposed an SSL framework for estimating the motion and depth of individual objects from monocular video. Their innovative approach models the motion of the object as a

rigid body with 6 degrees of freedom. It leverages keyframes within the video to generate depth maps, thereby achieving precise depth estimation. This model produces 3D descriptions of object movement in space and does not require external annotations for training. Experimental results on the KITTI driving dataset demonstrate the effectiveness of the SSL method in handling unlabelled video data and capturing object motion. Furthermore, the authors suggest that there is potential value in exploring the integration of scale information from external sources, as well as dissecting object motion into smaller components and employing different transformation techniques to enhance the model's capabilities further.

Spurr et al. (2021) and his colleagues are motivated by recent accomplishments of contrastive learning in model classification tasks, and introduced a new self-supervised 3D hand pose estimation approach. The approach utilised a self-supervised contrastive learning technique, training on an extensive set of unlabelled 3D hand pose image pairs and fine-tuning with only a small amount of labelled data. To gauge the effectiveness of their proposed method, the researchers conducted evaluations using the FreiHAND and YouTube3DHands datasets. Their results demonstrated superior accuracy when compared to existing state-of-the-art supervised methods, all within the same structured regression task.

According to Cheng et al. (2021), previous research on object motion recognition has focused on improving spatial representations of the human body while not fully utilising the temporal dependencies in action sequences. To tackle this issue, they developed a transformer-based model called Motion-Transformer, which leverages the self-supervised pre-training method to train with a large-scale video dataset to capture these temporal dependencies. Their experiments show that their proposed model outperforms previous state-of-the-art CNN-based methods in this area, demonstrating the effectiveness of the pre-training and fine-tuning paradigm.

Ma, Li and Li (2023) stressed an open problem of lacking labelled 3D human pose datasets. To solve this problem, they developed a novel self-supervised approach for training a 3D human pose estimation network using unlabelled data. Their model incorporates a specially designed factorisation network to separately predict the coefficients of canonical 3D human pose and camera viewpoint. Furthermore, they extended the application of the model to 2D poses, and the evaluation results demonstrated that the use of the SSL method maximally disentangled 3D human shapes and enabled accurate prediction of 3D human poses.

Huang, Fu et al. (2021) investigated a challenging task of infant motion analysis, to detect unique poses and actions of infants. Due to two primary factors of the infant data: high data scarcity limitation and the significant differences in body proportions between infants and adults, they developed an infant pose dataset comprising real-world infant images and synthetic infant poses to facilitate training. Moreover, they adopted a multi-stage TL-based model using an invariant representation learning strategy to transfer knowledge from the adult pose domain to the synthetic infant images. As a result, their pose estimation model achieved a mean average precision of 93.6, which effectively underscores the advantages of employing the TL to confront high data scarcity challenges.

Baur et al. (2021) focused their research on 3D scene flow in point clouds. Unlike prior research approaches, they adopted a novel perspective by training their model to prioritise stationary points over those in motion. They proposed an SSL approach by creating a self-supervised motion segmentation signal and then used it to predict motion information and identify stationary points. This information was then aggregated in the static parts of the scene to improve the performance of their model. The results obtained through their proposed method were highly encouraging. Across multiple real-world datasets, their experiment demonstrated state-of-the-art performance. Furthermore, the model exhibited robustness and generalisability, reaffirming the effectiveness of its approach.

Huang, Zhang et al. (2021) mentioned that motion information in videos is represented by the movement of pixels, which is challenging to label and use for training models with good motion representations. To address this issue, they proposed a self-supervised motion learning model called Motion from Static Images (MoSI), which learns motion from static images and encodes motion information by identifying significant motion regions and classifying pseudo motions generated by the model. The results obtained through MoSI demonstrated a substantial improvement in model performance, particularly in complex scenes and motions. This proposed model uses purely pseudo-labels instead of manual supervision to extract motion features, and the successful outcome underscores the advantages of the SSL methods in feature extraction and enhancing a model's transferability.

Bi et al. (2023) introduced a new SSL approach that incorporates enhanced spatiotemporal consistency for action recognition. Their method focuses on extracting motion information from video frames and incorporating static frames to create distracting video-positive samples, thereby reducing the impact of irrelevant variables on model discrimination. To differentiate between original frames and video frames, data augmentation techniques were employed to generate additional categories of negative samples to differentiate between original and video frames. The experimental results demonstrate improved performance and higher accuracy in recognising actions across various video action datasets.

In the motion field, SSL has gained more attention due to the difficulty, high cost, and inefficiencies associated with traditional manual annotation methods for point clouds, motion videos, and 3D modelling image data. One main advantage of the SSL method is that it has shown competitive performance in handling complex learning tasks, as exemplified by previous work (Cheng et al., 2021; Li et al., 2020; Spurr et al., 2021). Another advantage of the SSL method is to alleviate the data scarcity problem (Huang, Fu et al., 2021) and improve the model's feature extraction capabilities (Huang, Zhang et al., 2021). However, limitations are mentioned in current studies. One limitation is that general SSL pretext tasks like inpainting tend to promote invariant feature learning (Huang, Fu et al., 2021), which may not align well with downstream motion tasks such as estimation tasks. Moreover, Huang, Zhang et al. (2021) mentioned a gap in his study, where a commonly used motion learning strategy for transforming video data into static image data might cause time series information to be lost, and the significant movements in the background may also confuse the model.

6.2.6. Applications in environmental field

The environmental field encompasses various learning tasks, such as detecting water quality in geological settings, classifying animals and plants, and performing garbage classification in urban environments. While a large amount of unlabelled data is available for these tasks, the lack of well-annotated data is a standard limitation in this field. This is mainly due to the difficulty of collecting, storing, and annotating data. For example, geological detection and leaf disease recognition require expert annotation, which can be time consuming and expensive. Furthermore, the firm heterogeneity of real-world scenarios can lead to changes in the feature-space distribution of acquired data, requiring additional reconstruction when using traditional machine-learning approaches (see Fig. 22).

Cao and Xiang (2020) applied the TL method in garbage classification tasks by leveraging a pre-trained InceptionV3 model with Imagenet weights. This pre-trained model was then fine-tuned using a dataset comprising 20,000 images of common garbage, tailored specifically for downstream garbage identification tasks. To enhance the performance of the model, the researchers increased the size of the data through data augmentation and fine-tuned the pre-trained model. The results of their efforts were highly encouraging, showcasing the ability of the model to achieve high accuracy in the identification of trash in the real world. However, it is worth noting that certain factors, such as the quality of the target images, were not comprehensively examined in this study.

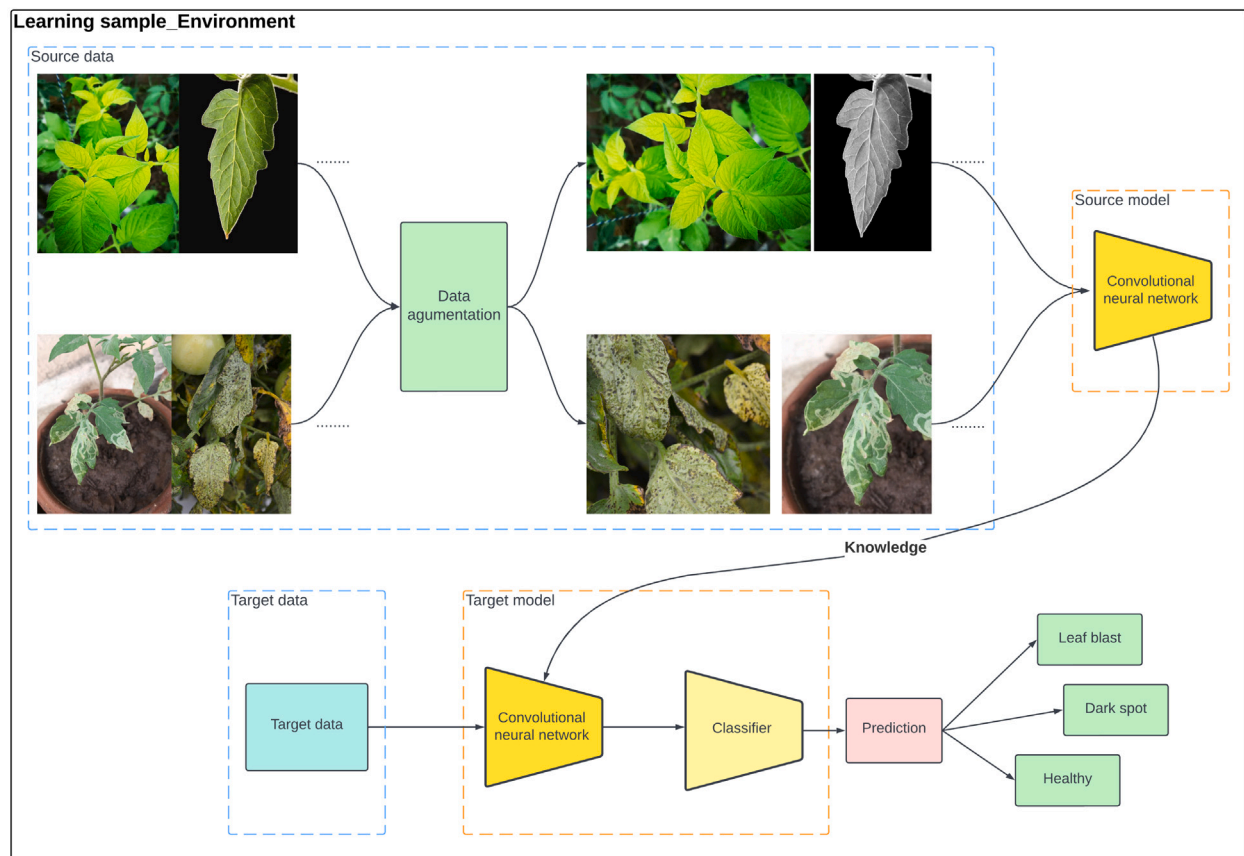


Fig. 22. Plant leaf disease diagnosis sample.

This leaves room for future research to explore and potentially improve these aspects.

Choe et al. (2020) conducted research aimed at creating a mobile application that can accurately classify endangered parrot species. Their approach involved the use of several neural network architectures, including ResNet50, InceptionV3, NASNetMobile and Inception ResNetv2, as the backbone encoders of their model. In their comprehensive study, they conducted a comparative analysis of model performance, contrasting the results of using ImageNet pre-trained weights with the results of training the model from scratch. In this framework, the neural networks of the encoder served as feature extractors, responsible for capturing key representations of parrots. The experimental results demonstrated the advantage of using the TL method, with an overall accuracy of 94.125% for the camera of the mobile device, certain species reaching 100% accuracy, and a recognition waiting time of only 455 ms.

Rehman et al. (2021) presented a parallel framework for real-time identification and classification of apple leaf diseases. They initiated their methodology by employing a pre-trained ResNet50 model with ImageNet weights for feature extraction. To tailor the model for the specific task of apple leaf disease classification, they conducted fine-tuning by replacing the last layer with a new one. This enabled them to retrain the model specifically for the identification of four target apple diseases. In addition to this, the researchers incorporated a masked RCNN model into their framework. This component was instrumental in creating feature maps and conducting segmentation tasks to identify various changes in plant leaves. The experimental results showed that their proposed method achieved an accuracy of 96.6%, which illustrates the superiority of the proposed ensemble model.

Huang et al. (2022) harnessed the TL method in tomato pests classification task using pre-trained CNN models. They opted for the

utilisation of four distinct neural networks, specifically AlexNet, InceptionV3, VGG16, and ResNet50, as the foundations for their investigation. The researchers compared the performance of these pre-trained models with other CNN + ML models such as SVM, KNN, and discriminant analysis, and found that the pre-trained ResNet50 model achieved the highest classification accuracy. All the models were tested under a relatively small dataset comprising fewer than 1000 annotated images that were collected by the research team from previous datasets and web searches. The results indicated several advantages of the TL method: simplify the feature extraction process, reduce training time, and provide faster assistance to users in real-world scenarios.

Attallah (2023) also conducted a study focusing on tomato plants. The research team utilised TL to train a neural network for the automated identification of tomato leaf diseases. By leveraging pre-trained CNN models, the target model obtained a more compact and high-level representation from the fully connected layer. The evaluation results demonstrated an impressive accuracy of 99.92% and confirmed the competitive performance of the TL-based CNN model in tomato leaf classification tasks.

Palanisamy et al. (2020) employed deep CNN architectures pre-trained on the natural image dataset as a strong baseline for environmental audio classification, marking a pioneering effort in tackling audio classification tasks using models initially designed for image data. They experimented with several well-established CNN models with ImageNet weight, including ResNet, DenseNet, and Inception, to assess their suitability for environmental and musical audio classification. To adapt the audio data for use with these image-based models, they processed the audio data by segmenting it into fixed-size windows and maintaining a consistent hop length across three channels to match the models' expected input format. As mentioned above, this is the first study to try to use an image pre-training model to handle the audio classification task, and despite the significant differences between audio

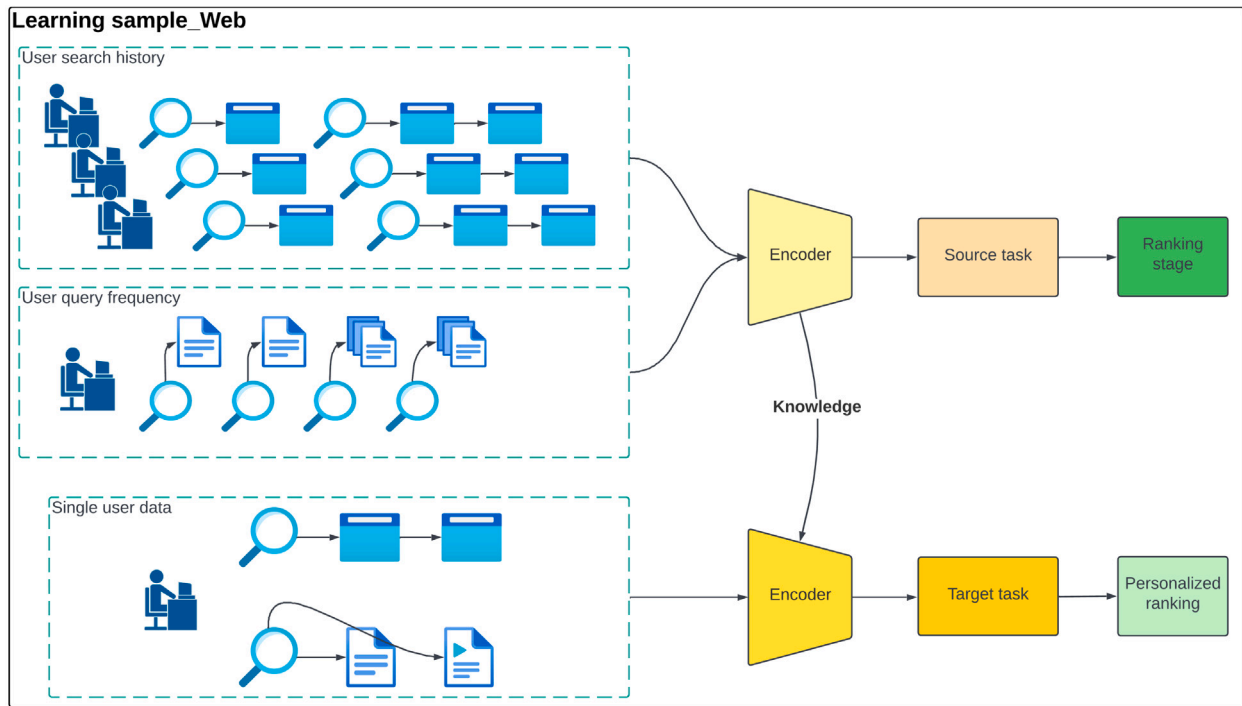


Fig. 23. User recommendation system sample.

spectrograms and standard ImageNet image samples, the experiments demonstrated that the model achieved a high validation accuracy of 92.89% within the TL setting and proved that different modality data can also lead to successful knowledge transfer and bring advantages to the model.

Chen et al. (2021) addressed the challenge of low data quality of water monitoring, where collected data often contain missing values due to human errors or system failures. They proposed a novel algorithm that leverages an unsupervised LSTM model and an instance-based TL model called TrAdaBoost-LSTM. The ensemble model combines the strengths of the LSTM model and TL techniques, learning from inherent long-term trend characteristics from the complete time series data and transferring it to the incomplete target data to predict the missing part. The result demonstrates a significant improvement of 15% to 25% in imputation accuracy compared to the baseline model.

Based on the aforementioned research, the TL method has demonstrated advantages in superior performance and faster response speed with applications (Attallah, 2023; Rehman et al., 2021), and also shows effectiveness in feature extractions (Choe et al., 2020). Additionally, the TL method has proven itself to be a valuable tool for mitigating the challenges posed by data scarcity limitations, which is essential for the practical implementation of AI applications (Huang et al., 2022). Notably, works have turned out that knowledge transfer between different modalities can still bring benefits to feature learning, although the specific principles and limits need further study (Palanisamy et al., 2020). It is important to acknowledge that a common issue prevalent in this field is the low quality of data. Data inaccuracy can stem from natural sources of noise, human errors during data collection or annotation, and technical issues related to data storage (Cao & Xiang, 2020; Chen et al., 2021). These factors collectively contribute to data noises, which can pose challenges when employing TL techniques that may struggle to effectively handle noisy data. Despite these challenges, the studies aforementioned in this field have all underscored the significant potential and benefits of TL in addressing real-world problems, offering valuable insights for the development and deployment of AI applications across various domains.

6.2.7. Applications in web field

The web field aims to provide virtual assistance towards social media applications. With the fast-growing web content products, more and more social media applications, platforms, and websites appear in people's lives. Within the past few years, neural networks have been successfully used to help service providers learn customer preferences and improve their services. However, due to the fierce competition between web companies, internet content has accelerated its update speed, which results in existing detection methods being outdated and failing to capture emerging knowledge. To against the changing scenario, we observed a brunch of new research for applying pretraining techniques in the web field, especially with the SSL method (see Fig. 23).

Shen et al. (2020) aimed to create a dynamic taxonomy system to manage the rapidly increasing volume of data in the field. Specifically, the goal was to provide real-time recommendations for online retailers and search engines on the Web. To achieve this, the researchers introduced a new self-supervised framework called TaxoExpan, leveraging taxonomy data to automatically generate pairs of taxonomy concepts, which were then used as training data to teach the model to predict whether a query concept was a direct hyponym of an anchor concept. Extensive experiments showed that TaxoExpan is a robust training objective that can reduce the model's sensitivity to label noise with the self-supervision data and effectively expand the taxonomy system without additional annotations.

Yao et al. (2021) proposed a multi-task SSL framework to construct a large-scale recommender system. The primary objective of the framework is to improve the understanding of the latent relationships between item features. Additionally, the authors devised a novel data augmentation method that leverages feature correlations within the framework. The evaluation results clearly indicated that the SSL model exceeded state-of-the-art regularisation techniques. This outcome underscored the advantage of employing self-supervised learning methods for enhancing model performance, particularly in scenarios where manual supervision data is lacking. These findings emphasise the practical utility of SSL approaches in real-world business applications.

Zhang et al. (2021) developed a self-supervised hypergraph learning framework for group recommendation. The challenge of this task involves complex high-order interactions both within and beyond groups, as well as the issue of data sparsity stemming from sparse group-item interactions. The proposed framework was designed to capture intra- and inter-group interactions among users and utilises self-supervised signals to regularise user representation, alleviating data sparsity issues. The results demonstrate the effectiveness of the proposed approach in improving group recommendation performance and the authors have recommended using the SSL method when designing group recommendation models to improve the recommendation performance.

Zhou et al. (2021) identified the problem of insufficient data in personalised search, where the user model requires a considerable amount of data to extract personalised signals. One of the primary challenges they encountered was striking a balance between learning high quality user representations and focusing on the final ranking quality, which is essential to improve model performance while maintaining model generalisability. Additionally, they observed that data sparsity was a widespread issue in existing methods, hindering the model's ability to learn powerful customer features. To overcome this challenge, they proposed a contrastive-based self-supervised sampling method that leverages paired self-supervised information from sequences of user behaviours in query logs. The experiments revealed that the proposed approach achieved state-of-the-art performance compared to the baseline models, but also managed to preserve the model's generalisability.

Feng et al. (2021) aimed to improve the current ML model's bot detection ability on Twitter. On the basis of observation from previous studies, their approaches had limited generalisation ability due to their reliance on a small amount of user information and training with few types of bots. To address these limitations, they proposed a self-supervised representation learning framework that jointly leverages user semantics, properties, and neighbourhood information. In this SSL approach, the model can effectively utilise a large volume of unlabelled information specific to each user. User behaviour is categorised into several distinct categories to create embeddings, which are then used to improve the final bot detection performance. The results of the experiment show that their proposed model outperformed the baseline bot detection models using the ML and deep learning method and demonstrated better generalisation ability in real-world scenarios.

Ameer et al. (2023) emphasised the challenge of classifying emotions from the huge amount of content generated by social media. Extracting emotions from short text segments can be a difficult task as a single short text line may contain multiple emotions, requiring multi-label classification. Previous works mainly relied on CNNs and did not explore the potential of multi-attention approaches such as Transformer and Bert. To address this gap, the research team employed LSTM and Transformer models with multi-attention layers to tackle this multi-label task. Their approach involved utilising emotion labels and emotion-causing characteristics to extract word embeddings. They then employed transfer learning to transfer the pre-trained embeddings to the target model. The results showcased the superior performance of the transfer learning multiple attention-based models compared to existing state-of-the-art ML approaches. However, they noted a limitation in the model's ability that failed to learn relationships between phrases and classes, and they suggested a future direction involving the use of various tokens to provide additional information for learning semantic representations.

Arbane et al. (2023) conducted a study to address the problem of the growing number of health-related discussions on social media platforms. Their research introduced a novel model utilising bidirectional LSTM for sentiment classification and analysing COVID-19 public opinions. Training data was collected from real-world platforms like Twitter. The experimental results demonstrated enhanced performance compared to recent conventional LSTM models, and the proposed

model can potentially provide valuable information for public health decision-making.

After reviewing several SSL approaches, it is evident that contrastive learning methods are popular in this field due to their effectiveness in learning representations from text data and in customer information retrieval. According to Shen et al. (2020), a significant advantage of SSL techniques in this field is reducing the need for additional human annotation and can be retrained dynamically to meet evolving customer requirements. Furthermore, Ameer et al. (2023) and Yao et al. (2021) demonstrated that SSL could improve item representation and generalisation learning as an additional regularisation technique. However, Zhang et al. (2021) expressed concerns about the lack of well-designed pretext tasks in their applications and highlighted the importance of designing general auxiliary pretext tasks. Feng's (Feng et al., 2021) ablation study also revealed that removing any aspect of information from the pre-trained model would result in a significant loss of performance, which researchers should pay attention to in future studies.

6.2.8. Applications in speaker field

The field of speaker-related research is dedicated to tackling tasks related to the localisation, verification, and classification of target speakers. Detecting speakers from audio data is commonplace for humans but still challenging for machines, as it requires the model to learn powerful representations that can effectively differentiate from the voices of human or non-human speakers. Unfortunately, training such models requires a large amount of labelled data, which is often scarce, as there is an abundance of unlabelled data available that cannot be immediately annotated and used in model training (see Fig. 24).

Huh et al. (2020) aimed to train an unsupervised speaker recognition model without relying on speaker labels. To achieve this goal, they proposed a self-supervised adversarial augmentation training strategy to learn speaker information. The method addresses the challenge of distinguishing speaker information from channel information using an augmentation classifier and gradient reversal layer. In more detail, their model utilised augmentation techniques to increase the size of the training data and trained a discriminator to classify inputs originating from the same channel while minimising binary cross-entropy loss. Experimental results showed that the proposed approach achieved state-of-the-art performance in self-supervised speaker recognition on the VoxCeleb and VOICES datasets.

Chi and colleagues presented a lightweight version of a self-supervised speech representation model named Audio ALBERT in their study (Chi et al., 2021). The proposed model used a pre-trained Mockingjay architecture and serves as a representation extractor for two downstream tasks: speaker classification and phoneme classification. During the pre-training stage, the model takes mel-spectrogram features as input and employs a masking strategy. Parts of the mel-spectrogram features are masked, and the model is trained to reconstruct them as a pretext task. This is a common strategy in self-supervised learning to encourage the model to learn useful representations from the input data. The authors also implemented a weight sharing technique in which the attention weights of previous layers are reused in adjacent layers to achieve faster inference and reduce the overall parameter size of the model without sacrificing performance. The results show that despite having significantly fewer parameters (91% less) than massive pre-trained networks, the proposed Audio ALBERT model achieved comparable performance.

In their study, Xia et al. (2021) examined self-supervised representation learning for speaker verification, comparing the performance of the SimCLR and MoCo methods. They found that the MoCo framework was superior in terms of learning speaker embeddings. Additionally, the researchers employed various augmentation strategies to evaluate their effectiveness in reducing extrinsic speaker variabilities between two random segments of the same speech utterance. The results showed

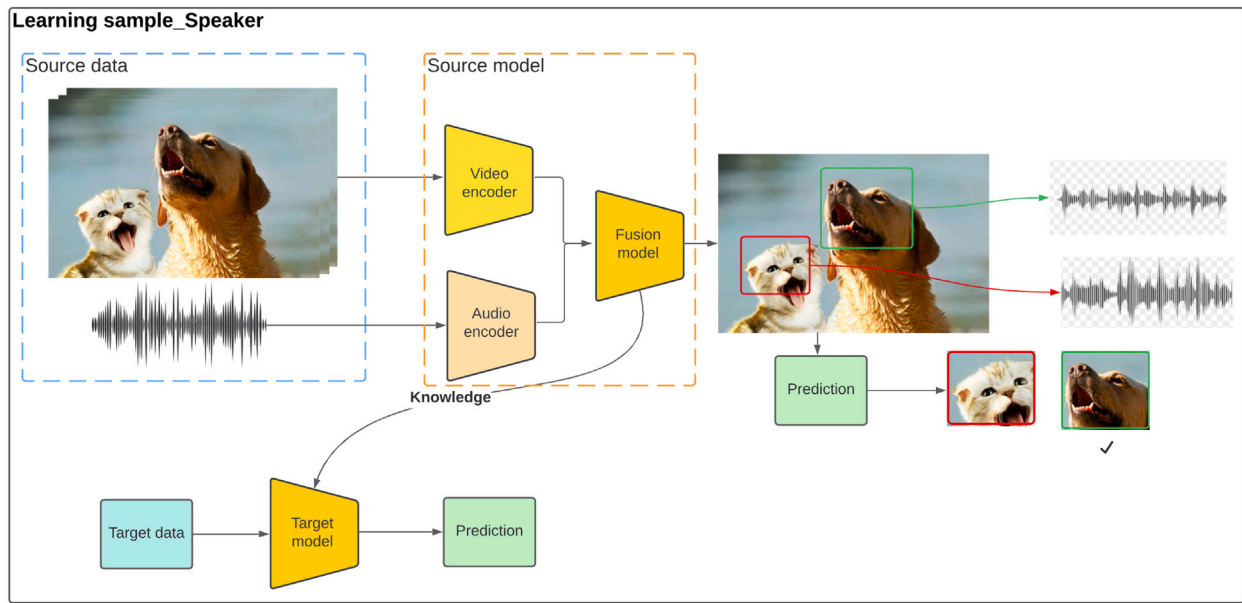


Fig. 24. Speaker recognition sample.

that augmentation significantly improved speaker representations for speaker verification tasks.

Sang and his colleagues proposed a novel regularisation strategy for self-supervised speaker representation learning to improve the performance of speaker verification systems (Sang et al., 2022). Their approach focuses on learning the similarity between latent representations of positive data pairs without using negative pairs. They used a Siamese network to implement their proposed architecture and through experiments, it demonstrated a big improvement in enhancing the performance of speaker verification.

Chen, Kanda et al. (2023) delved into applying SSL-based methods in Speech Separation tasks, building upon the foundation laid by previous SSL approaches like WavLM. Their approach involved pre-training a WavLM model using 300K h of unlabelled speech data and fine-tuning the pre-trained network by incorporating audio spectrogram concatenation with 10K h of data. A significant aspect of their approach was the enlargement of the dataset used for pre-training and fine-tuning. By incorporating a vast amount of unlabelled data, they aimed to enhance the model's ability to generalise and learn better representations. The experimental results showed promising outcomes in reducing the WER by approximately 10% across various test datasets. Moreover, the approach also reduced the 38% computational cost associated with downstream task learning, which is important for real-world applications.

Cai and his team introduced an iterative framework for self-supervised speaker representation learning using a deep neural network in their study (Cai et al., 2021). The framework involves an initial round of contrastive SSL, where the agreement between the different segments is maximised. The framework then uses a clustering algorithm to generate pseudo-labels and conduct more rounds of contrastive learning steps to minimise the contrastive loss. The network is trained continuously with each iteration, using pseudo-labels from the previous step to enhance its discriminative capabilities further. The experimental results demonstrated that the proposed iterative framework improved the performance of the speaker verification system compared to a purely contrastive self-supervised learning model. However, it is worth noting that each round of the iterative framework increases the training time as the training data size expands with each iteration.

Huang et al. (2023) aimed to tackle the problem of adapting speech recognition models from single-talker settings to multi-talker scenarios, where multiple speakers are present in the audio mixture. Such

scenarios pose challenges due to domain differences and the presence of multiple speakers. To address this challenge, they introduced a novel joint speaker modelling approach to leverage speaker embeddings to extract information from all speakers in the audio mixture. The key innovation is the use of the SSL method to train a model to learn generalisable task-agnostic representations. The experimental results demonstrated that the incorporation of speaker embeddings led to significant improvements in WER compared to a strong baseline WavLM system, which indicates that the speaker embeddings obtained from the SSL model are strong enough to enhance multi-talker speech recognition performance and highlights the efficacy of their proposed approach in multi-talker automatic speech recognition tasks.

Hu, Qian et al. (2020) have focused on a multi-modality object localisation task that uses parallel audio and video data to locate the speaker in video using the corresponding audio data. They introduced a method to learn reliable object representations from video data, which uses the SSL method to create class-specific audio object localisation. The approach aggregates the potential sound localisation outcomes in single-source scenes and generates an object localisation map based on preexisting object knowledge. The sounding objects are chosen to match the audio and visual object category distributions. The experimental results indicate that the proposed model performs well in filtering out silent objects and identifying the locations of sounding objects belonging to different categories.

Song et al. (2022) also focused on the task of sound source localisation and they proposed a self-supervised framework for predictive learning using explicit positive mining. To enable the model to focus on the target objects progressively and reduce the difficulty of positive pair learning, they designed a novel predictive coding module for audio-visual feature alignment. Experimental results have shown that their proposed approach can achieve significant improvements of 8.6% IoU and 3.4% AUocalization performance.

Despite significant advancements made in SSL approaches, researchers continue to design new strategies and methods. The SSL method has proven to be a simple but effective way to study audio tasks (Sang et al., 2022; Xia et al., 2021), and has demonstrated improved performance in downstream tasks while requiring fewer annotations (Huh et al., 2020). Furthermore, work from Chen, Kanda et al. (2023) and Chi et al. (2021) also proved the advantage of SSL methods in reducing the size of the target dataset and accelerating the training speed of the downstream task, which is significantly useful

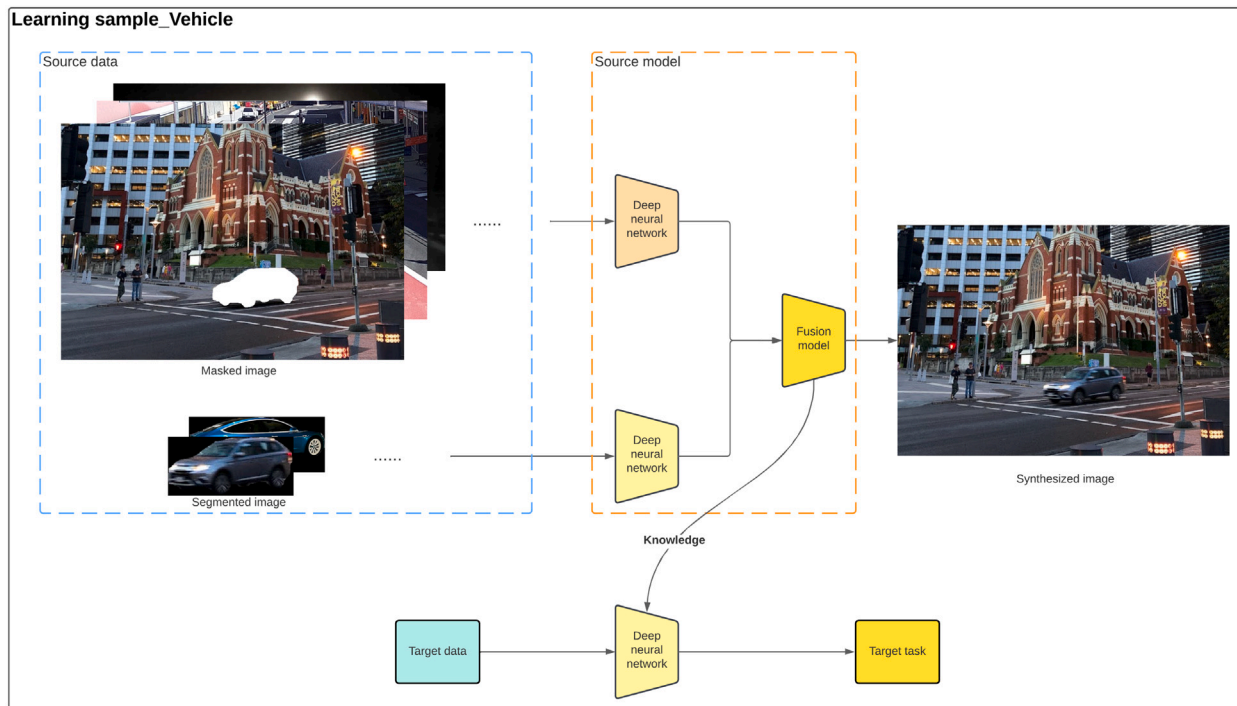


Fig. 25. Vehicle prediction sample.

when applied to real-world applications. Hu, Qian et al. (2020) and Song et al. (2022) shows a good result for utilising the SSL method in visual-audio multi-modality learning tasks, supporting the effectiveness of SSL approaches towards multi-modal learning. However, Cai et al. (2021) noted that one issue with pseudo-label generation is the presence of massive label noise, which requires prior data purification to maintain data reliability and reduce the computation cost involved in the data processing.

6.2.9. Applications in vehicle field

In the field of vehicles, machine learning techniques have been utilised to perform supporting tasks such as distance estimation and road segmentation. Unlike professional areas such as the medical field, acquiring training data in this field is relatively easy since modern vehicles come equipped with cameras to record video or image data. These vehicle and road monitoring data records can provide models with information about objects, structures, depth, and motion organised in a temporal sequence. However, despite this wealth of data, collecting large numbers of labelled data using manual annotations is still costly and challenging, especially in supervised learning settings (see Fig. 25).

Li et al. (2021) pointed out that previous research extensively focused on collecting various supervision signals, which required a significant amount of human labour. In response to this challenge, the authors introduced a novel self-supervised framework consisting of three branches. These branches include a global branch responsible for acquiring global codes from the image, an SSL branch dedicated to self-supervised representation learning, and a geometric feature branch focused on learning discriminative features. These three branches collaborate, offering the model valuable features. This work was also claimed to be the first SSL approach in geometric feature learning. The authors conducted a comprehensive ablation study to validate the efficacy of their proposed framework. The results clearly illustrated that the utilisation of self-supervised methods led to improved performance and enhanced the model's generalisation capabilities. Furthermore, it significantly reduced the need for manual data annotation efforts.

Kothandaraman et al. (2021) have introduced a new self-supervised method for road segmentation in adverse weather conditions. In this

approach, they have employed several techniques, such as generated pseudo-labelled labels, self-attention, entropy minimisation, and model distillation, to address the challenges faced in unsupervised learning and to enhance the model's accuracy. In the initial pre-trained phase, the SSL model adopts a self-attention auto-encoder architecture for training, using a dataset featuring clear weather conditions to capture essential road-related knowledge. Subsequently, in the downstream task, the authors improve the learning of the model by providing a limited amount of labelled data from the target domain, facilitating the handling of heterogeneous data. The experimental results have demonstrated that their method outperformed all previous unsupervised road segmentation and achieved a similar accuracy performance compared to previous supervised methods.

Kumar et al. (2021) identified a drawback in previous SSL approaches for monocular depth estimation, namely the scale ambiguity. This problem arises when the model struggles to generalise effectively when estimating distances in complex projection environments. To overcome this challenge, they presented a multitask learning approach that incorporates a self-attention-based encoder with robust semantic feature guidance for the decoder. In addition, they introduced a generalised robust loss function that eliminates the need for hyperparameter tuning and enhances the model's performance. Subsequently, the authors evaluated their model using the KITTI dataset and conducted comparisons with existing state-of-the-art monocular depth estimation methods. The experimental results not only establish their proposed approach as the new state-of-the-art design in this domain but also underscore the effectiveness of the self-supervised method in distance estimation.

Hu, Zhao et al. (2020) focuses on applying transfer learning to infrared pedestrian detection because this task is hindered by the lack of labelled infrared image datasets. To address this issue, Hu and his team propose a solution to pre-train the model in the ImageNet dataset and use a CNN model as the fixed feature extractor to learn representations from limited target data. The results demonstrate that pre-training on general images significantly improves the performance of the infrared model. This study shows potential for applying transfer learning to other infrared datasets and tasks by utilising existing large-scale nature dataset, and provides a future direction in developing

a domain-specific pre-trained model to further enhance the model's performance.

Farid et al. (2023) utilised the TL method in combination with the YOLO-v5 architecture to develop a robust model for vehicle detection and classification tasks in intelligent transportation systems. Their approach involved pre-training the YOLO-v5 model using massive labelled datasets comprising images and videos of traffic patterns. The pre-trained model was then fine-tuned using specific target data. Experimental results show that the fine-tuned YOLO-v5 model exhibited superior accuracy and execution time compared to other powerful baseline models.

Wei et al. (2023) applied SSL methods in depth estimation tasks to alleviate the need for labelled data and provide a cost-effective alternative to expensive depth sensors such as LiDAR. They introduced the SurroundDepth approach, which incorporates information from multiple surrounding views captured by cameras. The proposed model utilises a joint network to extract features from surrounding views and transfers them to a cross-view transformer to effectively fuse the learned information. Through extensive experiments, their models achieved state-of-the-art performance in multi-camera depth estimation challenges and accurately demonstrated the ability to predict real-world scales based on multi-camera extrinsic matrices.

Based on our observations of the proposed research, it is evident that recent SSL approaches have demonstrated exceptional performance in visual detection tasks related to the vehicle industry (Kothandaraman et al., 2021; Kumar et al., 2021). Through harnessing the capability to extract features from diverse sources of information, SSL approaches have demonstrated their superior effectiveness in acquiring both local and global representations, achieving comparable accuracy with supervised methods (Li et al., 2021). Moreover, Kothandaraman's work (Kothandaraman et al., 2021) has further validated that the SSL method is more resilient to challenging real-world scenarios than traditional unsupervised learning models, such as inclement weather, which can significantly advance the capabilities of the field. Notably, Hu's work has reminded researchers that the knowledge transfer between similar objects may help the model ignore the difference in colour channel.

6.2.10. Applications in robot field

The field of robotics encompasses a wide range of advanced technologies, from mechanical engineering to artificial intelligence. ML methods have been applied in this field for a considerable time, mainly due to the success of these approaches in computer vision tasks. However, the high cost of annotating real data for ML models, coupled with the limited ability of these models to generalise when visual differences arise, remains a significant challenge.

Yan et al. (2020) focused on the manipulation of the visual robot based on models of deformable linear objects. To address the high cost of annotating accurate data for ML models and the challenges of generalisation when visible differences arise, they proposed an SSL approach based on a state-space representation of the physical system. They used this information to build a dynamics model that could be learned in one setting and directly applied to other visually different settings. In particular, they demonstrated the first self-supervised training of rope state estimation on real images without requiring additional manual annotations.

In their work, Nubert et al. (2021) presented a self-supervised approach to LiDAR odometry estimation that can effectively use available LiDAR data and produce reliable estimates of robot poses. In this approach, the data collected by the LiDAR sensor is transformed into full point cloud scans, and the proposed network utilises these point cloud images to extract meaningful representations and construct a feature map. The proposed method does not rely on additional manual annotations, making it suitable for pose estimation tasks in scenarios where one has difficulty obtaining an accurate ground truth. Furthermore, the authors mentioned that in future work, they plan to

incorporate multimodal data, potentially combining it with temporal components to address local disturbances and enhance the model's robustness.

Ze et al. (2023) introduced a comprehensive framework that employs the SSL method to learn 3D representations for motor control. In this framework, they incorporated a pre-trained autoencoder, which learned from a substantial object-centric dataset through a pretext task. This task required the network to reconstruct a 2D view of the scene after applying a transformation to the deep voxel representation. The model effectively learned to encode geometric scene information and was subsequently fine-tuned using joint reinforcement learning on domain-specific data. Experimental results have shown enhanced sample efficiency in simulated manipulation tasks compared to approaches that rely on 2D representations. Moreover, the proposed framework can tackle robot motor control tasks in real-world scenarios.

The complex and unannotated nature of training data is a common limitation in the field of robotics, and from the above studies we can observe that self-supervised approaches have shown advantages in reducing models' dependence on manual annotations (Yan et al., 2020) and enhancing the efficiency in sample using (Ze et al., 2023). However, Nubert et al. (2021) identified a specified limitation when collecting real-world robot sensor data that rough terrains could introduce local disturbances and noise, which may require further research in this field to incorporate temporal components into network design.

6.2.11. Applications in physics field

We observed one work for the TL method in the physics field, which is a good practice for applying pre-trained methods for more application fields.

Inubushi and Goto (2020) introduces the application of transfer learning in non-linear dynamics tasks to solve fluid turbulence problems, enabling efficient predictions with limited data. The proposed transfer learning models learn from Lorenz chaos plot data from the source domain by predicting chaotic dynamics changes to provide an optimal transfer rate. Then, the model applied the learned parameters in the target domain for accelerating the nonlinear dynamic learning. The experimental results demonstrate the effectiveness of the TL method in chaotic dynamics tasks. In particular, this study is a new implementation of machine learning in the physics field.

7. Advantages and limitations of transfer learning and self-supervised learning

In the preceding section, we examined various research studies that implemented TL and SSL techniques in various domains, including medical, environmental, industrial, and others. Some of these studies drew on methods from computer vision tasks, while others utilised techniques from text mining and audio analysis fields. From our observation, TL has been more frequently employed in medical, industrial, NLP, and environmental domains. At the same time, SSL has gained more popularity in the remote sensing, motion, web, speaker, and vehicle domains. In this section, we highlight the advantages and limitations inferred from the aforementioned studies and group them to help readers understand better. In Table 3, we conclude a list of the advantages and limitations for comparison.

7.1. Advantages of transfer learning and self-supervised learning

Both TL and SSL methods contribute to the improvement of target models. While the two methods share some common advantages, each of them also has unique merits. Therefore, we classify the advantages of these two methods and explain their causes and their impact on the training process.

Table 3
Advantages & Limitations of TL and SSL.

Descriptions	Transfer learning	Self-supervised learning
List of advantages		
Alleviating data scarcity	✓	✓
Improving the model performance	✓	✓
Accelerating model's convergence speed	✓	✓
Improving model's generalisation capability	✓	✓
Making use of large-scale datasets	✓	✓
Preventing models from overfitting	✓	✓
Support models to learn from simulations	✓	✗
Support reducing labelling efforts	✗	✓
Support leveraging multi-modal data	✗	✓
Support dynamic and continuous training	✗	✓
List of limitations		
Risk of domain mismatch	✓	✓
Reliance on large-scale pre-training data	✓	✓
Lack of data quality handling capability	✓	✓
Lack of explainable capability	✓	✓
Risk of label shift	✓	✗
Risk of model focus shift	✓	✗
Lack of well-designed pretext tasks	✗	✓
Lack of powerful supervision signals	✗	✓
Substantial computational cost	✗	✓
Risk of low pseudo-labelling reliability	✗	✓

7.1.1. Shared advantages of transfer learning and self-supervised learning

In this section, we summarised five shared advantages of TL and SSL methods, including reducing the reliance on labelled data, improving target model learning, accelerating convergence speed, strong generalisation capability, and publicly accessible training datasets.

One primary advantage of TL and SSL methods is their ability to reduce the reliance on large-scale annotated data. As previously mentioned, achieving high-performance deep learning models often necessitates vast datasets. From our observations, data scarcity poses a significant challenge across various domains (Akrim et al., 2023; Huang, Fu et al., 2021; Xiao et al., 2023), potentially leading to overfitting, performance degradation, and hindrance in the advancement of deep learning applications. Our review reveals the existence of numerous TL and SSL methods explicitly tailored to address this issue. These methods leverage either existing extensive datasets or gather new data to train models and transfer knowledge in the form of learned parameters to the target domain. Omran et al. (2023) and Palanisamy et al. (2020). Using shared knowledge, the target domain starts with a better foundation compared to training from scratch, which requires only a small amount of annotated data to fine-tune (Akrim et al., 2023; Spurr et al., 2021). Notably, the effort to collect and annotate large-scale data for deep learning applications has proven to be financially and labour-intensive. For instance, the widely-used ImageNet dataset, which consists of 14 million annotated images, has cost three thousand Mexican workers months and millions of dollars to annotate. The situation gets even worse in professional domains like medicine and mechanics, with up to 30 times higher per-image annotation expenses. Consequently, the advantage of TL and SSL methods in diminishing reliance on large-scale data proves indispensable when applying deep learning to real-world applications.

The second advantage of both TL and SSL methods is that they can boost the target model learning and improve target model performance. This enhancement is achieved through positive knowledge transfer, allowing the target model to construct networks with deeper layers and more parameters without the need for an extensive amount of training data. Furthermore, TL and SSL approaches contribute to the creation of a superior feature extractor by utilising pre-trained networks with valuable and potent features, thereby enhancing data efficiency and generalisation capabilities. The experimental results demonstrate that both TL and SSL methods have shown remarkable outcomes in

improving model performance (Li et al., 2021; Song et al., 2022). Experiments also proved that fine-tuning with large pre-trained models can perform better than training from scratch with limited available data (Choe et al., 2020; Moor et al., 2023).

The third frequently mentioned advantage of TL and SSL methods is that they can accelerate the convergence speed in target model training. This is a widely recognised benefit offered by pretraining techniques because the target model can use the transferred parameters from the source domain directly and fine-tune them with fewer target data instead of training from scratch. As observed across numerous studies (Chen, Kanda et al., 2023; Gomroki et al., 2023), this advantage significantly reduces the time required for the model to reach convergence. Additionally, it contributes to a reduction in computational costs, which holds practical value when applying these methods to real-world applications.

The fourth notable advantage of TL and SSL methods is their strong generalisability. TL and SSL approaches rely on extracting generalised features from one task to another. Using neural networks, these methods extract representative characteristics and compare them with new data to make decisions or generate new features. In addition, certain universal features have the ability to not only represent their domains but also be applicable in other related fields. This generalisation capability is significant in deep learning, as training a model often requires substantial time and computational resources, while building reusable models can help save a lot of resources and expenses. Researchers aim for their trained models to be applicable in a broader range of fields and scenarios, maximising the value and applicability of their efforts (Cao et al., 2020; Chen et al., 2022).

The fifth advantage of TL and SSL methods lies in the availability of publicly accessible large-scale training datasets. Although these methods are designed to reduce the dependency on data from the target domain, they still require substantial datasets for pre-training the models and ensuring the acquisition of comprehensive knowledge. Over the years, TL has witnessed significant progress, developing well-organised datasets such as ImageNet-21K with 14 million labelled images and COCO with 328K annotated images. This availability of such publicly accessible large-scale data has proven to be highly beneficial in numerous TL and SSL studies, as demonstrated in references like (Hu, Zhao et al., 2020; Rehman et al., 2021). Although in most fields, the proposed SSL approaches are relatively recent, these publicly accessible resources benefit SSL research by providing scientists with direct access to extensive datasets, eliminating the need to invest valuable resources in data collection and annotation. Additionally, the development of SSL approaches has broadened the scope of valuable unlabelled datasets, offering researchers new opportunities and choices for their work.

7.1.2. Distinct advantages of transfer learning

TL methods offer the advantage of enabling models to learn from simulations. These simulations provide a controlled and safe environment for training the model to perform specific actions. Training in a simulated setting can thoroughly test and evaluate the model's performance under various conditions and scenarios. In domains where safety is paramount, such as self-driving car systems (Farid et al., 2023), the use of simulations for training is particularly practical and beneficial. It allows the model to gain crucial experience and learn optimal actions without the risks associated with real-world testing. Once the model has achieved satisfactory performance in the simulation, the gained knowledge and insights can be transferred to real-world applications with greater confidence.

7.1.3. Distinct advantages of self-supervised learning

We conclude three advantages of SSL as follows: saving labelling efforts, effectively leveraging multimodal data, and facilitating dynamic training.

One key advantage of the SSL method lies in its remarkable reduction of labelling efforts. In traditional annotation workflows, human

experts must carefully analyse and understand the data before providing accurate labels or classifications. However, SSL approaches leverage large amounts of unlabelled data and unsupervised learning techniques to generate pseudo-labels through data augmentation instead of using manual labels. This results in a minimal requirement for manual annotation efforts, primarily during the following task training phase, as demonstrated in studies such as Kothandaraman et al. (2021) and Wei et al. (2023). In contrast, TL methods still necessitate a certain number of labels to support the training process, whether in the pre-training stage or fine-tuning stage. Moreover, recent advances in SSL have demonstrated performance comparable to supervised learning methods, highlighting the effectiveness of the generated labels and the immense potential of SSL in future research and applications (Huang, Zhang et al., 2021; Kothandaraman et al., 2021).

SSL approaches offer an additional advantage in effectively leveraging multimodal data. A prominent example is the CLIP model developed by OpenAI, which integrates image and text models into a single framework. By performing contrastive learning using the output of an image encoder and a text encoder, CLIP facilitates exploring relationships between different modalities. Similarly, as we have observed in our cited references, numerous research efforts have utilised state-of-the-art SSL architectures such as transformers and BERT to leverage data from multiple modalities, which has consistently led to improved performance and generalisability of models (Fedorov et al., 2021; Nubert et al., 2021). This advantage allows researchers to harness the corresponding features from different modalities, and build models that capture rich interactions between them and employ meaningful representations across diverse data types.

SSL approaches provide another advantage in their ability to facilitate dynamic training. Dynamic training, also known as continuous training, involves continuously updating and expanding an existing model to adapt to evolving environments. Different from traditional supervised learning models that require substantial time and effort to prepare labels for newly collected data, SSL approaches are particularly well-suited for dynamic learning due to their self-supervised nature. SSL methods empower the model to fine-tune its parameters using unlabelled data, thereby saving considerable resources and time that would otherwise be spent on data labelling. This advantage allows the model to keep learning and stay updated with changing trends and patterns. For example, in the context of social media platforms, where topics among users change rapidly, dynamic learning enables the model to continuously incorporate up-to-date knowledge (Shen et al., 2020). In addition, this advantage eliminates the need for labour-intensive annotation efforts and enables the model to adapt quickly to new information. Furthermore, SSL approaches possess strong learning capabilities compared to traditional unsupervised methods. They are able to extract more meaningful and useful features from training data, making them effective in capturing relevant patterns and insights (Yan et al., 2020). By enabling dynamic training, SSL approaches facilitate continuous learning and adapt from up-to-date data, ensuring that the model remains effective in dynamic and evolving environments.

7.2. Limitations of transfer learning and self-supervised learning

In addition to the advantages mentioned above, TL and SSL methods also have certain limitations. Some of these limitations are specific to certain fields or areas, while others are more general and commonly observed. We categorise these limitations, provide explanations for their occurrence, and highlight their impact to help researchers draw attention to their studies.

7.2.1. Shared limitations of transfer learning and self-supervised learning

In this part, we discuss four common limitations of TL and SSL, including domain dissimilarity, lack of pre-training data, low data quality, and lack of explainable techniques.

The first limitation we want to stress for both the TL and SSL methods is the domain mismatch risk. This problem is caused by the discrepancy or dissimilarity between the domain data or the divergence between their distributions. It is a crucial cause that led to negative transfer when pre-training models. This problem reduces the efficiency of the knowledge transfer process and harms the performance of target domain models. For computer vision learning tasks, the mismatch often occurs when there are fundamental differences between the source and target domains regarding their data's characteristics, including the number of channels, intensity, location, scale, and orientation. For instance, the widely used pre-training dataset ImageNet contains vast numbers of 2D natural RGB images. In contrast, data used in other professional fields such as medicine may only involve 2D greyscale images, 3D structured data, or video data, which include extra time series information (Palanisamy et al., 2020). Differences in colour, feature, and even pixel pose a high risk of domain mismatch between the source and target domain, leaving unique challenges in the development of effective computer vision models for professional fields (Chen et al., 2022; Xiao et al., 2023). Consequently, this limitation has placed pressure on researchers to develop and tailor suitable models, architectures, and techniques that can effectively address these variations when adapting pre-trained methods to new environments.

The second limitation of the two methods is their reliance on a large amount of pre-training data, which can be attributed to the domain mismatch issue. The need for a substantial pretraining dataset arises from the necessity of TL and SSL models to acquire more comprehensive features and knowledge (Feng et al., 2021), thus enhancing their ability to generalise to unknown samples. Despite the aforementioned availability of numerous large-scale datasets for TL and SSL methods due to years of development, not all of them are suitable for every domain. In professional domains such as industry, remote sensing, and medicine, the issue of data scarcity for pre-training persists and cannot be resolved by relying on large datasets like ImageNet due to the risk of introducing domain mismatch. Utilising small pre-training datasets may increase the risk of overfitting, reduce models' reliability when training the target model, and waste valuable time and resources. However, it is crucial to exercise caution when selecting and expanding the pre-training dataset to avoid domain mismatch (Chen, Kanda et al., 2023). Inappropriate choices and expansions can introduce noise and result in a negative transfer, undermining the effectiveness of the entire transfer process. This challenge may be particularly pronounced in data characterised by unique attributes, such as medical, satellite, and tabular data, which require further attention in future studies.

The third limitation that we want to highlight is the data quality problem. Different from the domain mismatch that occurred in many different application areas, the quality problem is mentioned more frequently in specific areas like environment and vehicle fields (Cao & Xiang, 2020; Chen et al., 2021). This is because they are susceptible to environmental issues, the data storage process, and human errors. Due to the nature of these two pre-training methods, the noise in the data may be transferred from the source domain to the target domain, affecting the credibility of transferred knowledge and leaving risk to the following users. Addressing the issue of low-quality data can involve implementing rigorous data screening and preprocessing techniques, but at the same time, they may increase the computational cost, project expense, and lead to other limitations.

The fourth limitation pertains to the lack of explanations of the TL and SSL methods. Currently, many machine learning models operate as "black-box" systems, where the decision-making process remains opaque to the user. Additionally, existing studies on ML explanations primarily target users with ML expertise, assuming a certain level of background knowledge and common sense. In the case of pre-training methods like TL and SSL, this challenge is exacerbated due to the inherent complexity of the two-stage training process. Consequently, there is a lack of clarity regarding how target models leverage transferred knowledge and the subsequent impact on final decisions. This limitation increases the likelihood of transfer failures and imposes additional time costs on users, thereby raising the barriers to accessing and utilising such technologies.

7.2.2. Distinct limitations of transfer learning

One limitation in the TL process is the occurrence of label space shifts. This refers to where the source and target domains differ in the available labels. Typically, researchers aim to avoid using dissimilar data for knowledge transfer (Chen et al., 2022), ensuring that both domains share similar labels. However, in some instances, such as distant domain transfer learning, where statistical-level close data, instead of visually comparative data, are used during pre-training (Palanisamy et al., 2020), the label shift problem may arise more frequently, limiting the model's performance and leading to negative transfer.

Another limitation of TL is the issue of model focus. Previous experiments in supervised learning have revealed that models often prioritise features that are not directly related to the target object of interest. For example, given a pre-trained model utilising ImageNet weights to analyse a picture of a bone X-ray image, the model may disproportionately focus on the irrelevant bone area instead of the real disease area, which is the intended focus. Resulting in the model making predictions based on less relevant elements rather than the expected target is disheartening. Recently, more and more studies in TL have raised the same concern (Ankenbrand et al., 2021). Moreover, due to the lack of explainability in TL, there have been relatively few investigations in this area. Researchers speculate that this problem arises because the model considers a broader data context. This limitation hampers the reliability of the TL technique and poses challenges for its real-world applications, calling for further attention and research from the scientific community (Albahri et al., 2023; Alzubaidi et al., 2023).

7.2.3. Distinct limitations of self-supervised learning

There are four limitations of SSL in this section, which are lack of strong supervision signals, lack of pretext tasks, huge computational costs, and potential label errors.

One limitation of SSL is the insufficient availability of well-designed pretext tasks. Currently, the most commonly used pretext tasks, such as predicting rotation, image inpainting, colourisation, jigsaw puzzles, and super-resolution, are primarily designed for computer vision tasks. However, a limited range of pretext tasks are available for video, sound, NLP, and other domains. This discrepancy is due in part to the popularity of computer vision tasks and the challenges involved in designing auxiliary tasks for less intuitive data types. We also observed several attempts in designing pretext tasks for the specific domain and downstream tasks (Li et al., 2023; Zhang et al., 2021), which underscores the current lack of well-designed pretext tasks in some domains. Pretext tasks play a crucial role in the SSL learning process and significantly impact the quality of the extracted features and representations. Thus, the lack of diverse pretext tasks may impede performance improvements and hinder further advancements in the field of SSL.

Another limitation of SSL is the lack of powerful supervision signals. Traditional supervised learning models rely on precise annotations provided by domain experts, while weakly supervised models use a feature extractor to learn representations from unlabelled training data. In contrast, SSL methods use generated pseudo-labelled labels to provide supervision signals (Zhang et al., 2021), bridging the gap between weak and strong supervision. However, in some studies, it has been observed that a lack of powerful supervision can hinder the training process and result in failure when dealing with complex tasks such as multiclass motion segmentation (Baur et al., 2021). Furthermore, due to the lack of strong manual supervision signals, there is another risk that the performance and reliability of the trained model may be compromised due to inappropriate supervision in SSL.

SSL is also subject to the substantial computational cost associated with pretext task learning. Based on observation, researchers often devise intricate pretext tasks for their models to acquire invariant representations in a diverse range of images and augmentation transforms (Li et al., 2023; Xue et al., 2020). These tasks enable the model to learn more valuable features from the training data. However, it has been

observed that these pretext tasks typically require significantly higher computational resources compared to their supervised counterparts. According to Koçyiğit, Hospedales, and Bilen (2023), reaching similar accuracy levels in pretext tasks requires approximately 23 times more computation, including 8.8 times more iterations and 2.6 times more computation per iteration. This heavy reliance on complex semantic transformations and extensive analyses necessitates the additional allocation of computational resources. As a result, project timelines may be extended, adjusting model parameters can become challenging, and overall project expenses may escalate.

The final limitation of SSL that could be highlighted is the challenge of the accuracy of generated labelling. As mentioned above, SSL methods rely predominantly on self-generated signals to supervise the training process, which introduces the risk of label inaccuracies. These inaccuracies can stem from issues in selecting appropriate loss functions or employing incorrect augmentation techniques for generating pseudo labels. This limitation may lead to error-prone tasks and detrimental effects when transferred to the target domain. Furthermore, as SSL approaches generate labels using their own models, it becomes difficult to identify and address these labelling errors effectively. Hence, researchers must exercise caution when selecting training data and tasks to mitigate the impact of labelling inaccuracies and ensure the reliability of the SSL approach.

8. Discussion

In this section, we conducted a performance comparison based on different domains, data types, and downstream tasks to show the effectiveness of the TL and SSL methods. Furthermore, we have provided a table of recommendations for choosing pre-trained methods based on different domains (referred to Table 4). Finally, we list several potential directions for future research of the TL and SSL methods, discussing the current gaps and the need to improve them in future work.

8.1. Performance comparison

To offer a more comprehensive understanding and substantiate the factors that influence the performance of pre-trained models, we have conducted a thorough performance comparison among the reviewed TL and SSL approaches. This comparison takes into account various key factors, including differences in application domains, downstream tasks, dataset sizes, data types, and training speed.

In our analysis of the image data, the TL and SSL methods have demonstrated distinct characteristics and advantages. In particular, both the TL and SSL approaches exhibit superior performance in various tasks, including image classification (Bargshady et al., 2022; Fedorov et al., 2021), image segmentation (Azizi et al., 2023; Zhao et al., 2021), highlighting their versatility. Moreover, both methodologies have made substantial strides in reducing data dependency, with the TL method effectively diminishing 90% of the required data within the target domain (Ankenbrand et al., 2021), and the SSL method can even maintain satisfactory performance with 1% of the available target dataset (Azizi et al., 2023). Regarding the performance of selected tasks, TL methods pre-trained on ImageNet display consistent performance across different image data with both colourful and grey-scaled image datasets (Ankenbrand et al., 2021; Xue et al., 2020). In comparison, the SSL method exhibits superior adaptability, successfully handling diverse image structures, including 2D medical images (Moor et al., 2023), 3D human images (Ma, Li and Li, 2023), point cloud images (Baur et al., 2021). Furthermore, both the TL and SSL approaches have shown effectiveness in utilising multimodal data, as the TL method (De Bois et al., 2021) has shown good model performance by producing valuable features. At the same time, SSL-related work has demonstrated more advantages such as leveraging in-distribution and out-of-distribution image data to improve model generalisability (Fedorov et al., 2021), and enhance model performance (Azizi et al., 2023).

Table 4
Recommendation of methods for different fields.

Application field	Recommended method	Explanation
Medical	Self-supervised learning	Ability to train with unlabelled data, non-sensitive towards colour information, and can significantly reduce label dependency in the target domain.
Industrial	Transfer learning	Can make use of existing large-scale dataset for pre-training and reduce application risks by performing simulations.
Remote sensing	Self-supervised learning	Alleviate data scarcity, reduce domain mismatch risk, and reduce labelling shift risks.
Natural language processing	Self-supervised learning	The ability to learn powerful representations from unlabelled data.
Motion	Self-supervised learning	Ability to learn intra-information from complex data.
Environment	Transfer learning	Can use existing large-scale datasets for pre-training.
Social media	Self-supervised learning	Can make easier use of dynamic learning and reduce labelling effort.
Speaker	Self-supervised learning	Reduce computational cost, reduce the influence of inaccurate annotations, and improve training speed.
Vehicle	Transfer learning	Can make use of existing large-scale dataset for pre-training and reduce application risks by performing simulations.
Robot	Self-supervised learning	Take advantage of abundant unlabelled data to learn more effective features.

In the realm of text data, TL methods offer a distinct advantage by significantly reducing data requirements, as evidenced by studies such as Elmadany et al. (2022). On the other hand, SSL methods have addressed the problem of data scarcity by effectively utilising unlabelled text data, as highlighted in Liu et al. (2020). When it comes to task performance, TL methods have proven their effectiveness in tasks such as text generation and question-answering, as demonstrated in studies such as Elmadany et al. (2022) and Sung et al. (2022). In contrast, SSL methods have shown satisfactory performance in challenging scenarios, including recommendation and chat generation tasks, which often rely heavily on large-scale pre-training data, as indicated by Liu et al. (2020) and Zhang et al. (2021). Moreover, SSL approaches bring additional advantages by providing valuable semantic embeddings and powerful text representations, as highlighted in study (Liu et al., 2020).

In the context of audio data, SSL methods have proven their effectiveness in various audio-based tasks, including audio classification (Sang et al., 2022), object localisation (Huang et al., 2023), and speech separation (Chen, Kanda et al., 2023). In particular, SSL methods have demonstrated clear advantages when combining video and audio data in training, providing additional visual information that accelerates audio localisation tasks, as indicated in Song et al. (2022). In contrast, TL methods have not shown the same level of efficiency in audio-based tasks, with only a limited number of studies conducted in earlier years. One notable work highlights the successful transfer from an image-based pre-training model to an audio-based target training scenario (Palanisamy et al., 2020).

Video data can be regarded as a sequence of images with extra time-series information. Based on our limited observations, SSL methods have shown advantages in learning from unlabelled video data and capturing valuable time-series information (Bi et al., 2023; Huang, Fu et al., 2021).

In summary, TL approaches demonstrate efficiency in image-based tasks including classification and segmentation, particularly in the fields of industrial, vehicle, and environmental. TL approaches have the capability to maintain function across limited data sizes, but having access to more pre-training and target data can enhance the performance of TL models and ensure their continued effectiveness. The most suitable domains for applying TL methods are those that possess or exhibit similar features to the current large-scale publicly available datasets, including domains related to environmental and vehicle care. In comparison, SSL approaches exhibit distinct advantages when handling diverse data types-related tasks including text, audio, and video-based tasks, showcasing superior generalisability and performance. Additionally, SSL methods are more flexible in terms of data size, as they

can maintain their effectiveness even with extremely small datasets. Furthermore, SSL models can achieve better performance and enhanced generalisation capability when trained with multimodal data related to the task. SSL methods are particularly well-suited for domains characterised by vast amounts of data that are challenging to annotate, like natural language processing and speaker domains.

8.2. Method recommendation

To better support the following researchers in choosing the pre-trained method, we want to give our recommendations based on the above-summarised advantages, limitations, and performance of transfer learning and self-supervised learning. Table 4 provides a clear explanation for each of the application fields.

We recommend the TL method for the industrial and vehicle fields because the data collected in these fields are more general and can take advantage of publicly available pre-training datasets like ImageNet. Additionally, the ability of the TL method to conduct simulations can help the model avoid potential risks, especially suitable for machine tests and diagnosis in the industrial field and self-driving tests in the vehicle field. The generalisability of the TL method is also an advantage in these fields, as they typically involve heterogeneous data.

The TL method is highly recommended for applications in the field of environment. Given the abundance of publicly available large-scale datasets, particularly with ImageNet, which comprises nature images. Choosing TL methods can easily leverage these resources and utilise well-trained pre-training models shared by the research community, which not only saves computational resources, but also significantly reduces training time. Current experiments also demonstrated that TL models exhibit advanced performance in environmental research and applications.

In the medical field, we recommend employing the SSL method due to its advantages in achieving satisfactory performance with limited data. Experiments have demonstrated that SSL models can achieve or exceed baseline models using only 1% of labelled data compared to training from scratch (Krishnan, Rajpurkar, & Topol, 2022). This significant improvement helps address the long-standing challenge of data scarcity in medical research. Furthermore, the use of SSL methods can contribute to reducing labelling effort, as annotating medical data requires the expertise of professionals and is a time-consuming process. Another notable advantage of SSL for its ability to use multi-modal data easily, as there are multi-types of data existing in the medical field, such as image, video, audio, and electronic records that can help accelerate

model training. Furthermore, studies have shown that self-supervised learning approaches tend to learn invariant features and ignore colour information (Ericsson et al., 2021), which could also be an advantage in the medical field, as most medical image data are greyscaled.

Regarding the field of remote sensing, we recommend the use of SSL methods for two primary reasons. The first reason is that remote sensing data, predominantly satellite images, possess distinct characteristics that differ significantly from commonly used natural image datasets like ImageNet. This disparity between domains increases the risk of domain mismatch when applying TL methods with ImageNet (Chen et al., 2022), which necessitates the collection of domain-specific data. On the contrary, SSL methods can leverage publicly available unlabelled satellite datasets to address data scarcity issues in remote sensing, saving time and costs (Gomroki et al., 2023). The second reason pertains to the complexity of satellite images. A single satellite image often contains numerous objects, such as rivers, trees, roads, and buildings, making manual labelling challenging, particularly in remote sensing-related classification and segmentation tasks. TL methods may be prone to label shift issues in such scenarios. However, SSL methods can use pseudo-labels to effectively learn powerful features from target data, mitigating the risk of label shift and yielding improved performance (Li et al., 2023).

We recommend the SSL method for the NLP field for its advantage of providing self-generated supervision signals. In contrast to traditional unsupervised learning methods, the adoption of SSL techniques allows models to acquire more powerful representations. Furthermore, the presence of substantial unlabelled text data in this domain provides SSL methods with the opportunity to expand their training dataset, leading to improved performance outcomes. Furthermore, the availability of well-designed self-supervised multi-modal approaches can help the model learn rich representations and enhance model performance, making it a valuable tool for NLP tasks.

In the motion field, we recommend the SSL method due to the distinct channel characteristics of motion data compared to natural images, which may cause a mismatch in knowledge when using the TL method. The prevalent data types include video, point clouds, and 3D images, all of which possess distinct representations and structures compared to the commonly encountered 2D natural datasets typically associated with TL approaches. This inherent dissimilarity between domain features makes it challenging to apply TL methods seamlessly across these diverse data types. In contrast, the SSL method shines by effectively harnessing a substantial volume of unlabelled data. It leverages the temporal sequencing information and intra-data relationships inherent in 3D structures and point-cloud data, resulting in the generation of more robust and powerful representations.

In the social media field, we recommend the SSL method for its ability to perform dynamic learning more easily and save labelling effort. As mentioned above, models in the social media field often require frequent updates in training data and knowledge, making methods such as SSL advantageous in reducing the cost of retraining and enabling updates without much expense in annotating data. Moreover, within the realm of social media data, the predominant data types revolve primarily around text-based content and user records. Annotating such data can be a time-consuming process, and SSL methods prove to be invaluable in alleviating this manual annotation burden, ultimately saving significant time and resources.

We recommend utilising the SSL method in the speaker field because this field has many unlabelled resources that can be used to learn representations, and sometimes human-annotated data in this area might also be inaccurate since human annotator may also make mistakes towards tasks like speaker verification when lacking additional visual information. The advantage of the SSL method in this field lies in its ability to obtain powerful representations from unlabelled data. Moreover, it excels in fusing multi-modal data from both visual and audio sources, thereby enriching the learned features and mitigating the impact of potentially inaccurate annotations. Furthermore, the

adoption of contrastive learning strategies and the availability of well-designed pretext tasks within SSL methods within this field empower models to learn discriminative representations directly. This not only saves computational resources, but also accelerates the training process compared to more general representation learning methods.

We recommend SSL for the robot fields because of the limited availability of labelled data in this domain. Compared to TL methods, SSL reduces the risk of domain mismatch by directly using unlabelled data from the target field. SSL method can also achieve better performance by leveraging information from multimodal data such as visual and audio to help the model achieve better performance.

8.3. Future work

This section provides a summary of the future directions highlighted in the reviewed studies and offers our own insights on potential areas for future work, taking into account the existing gaps in TL and SSL applications.

Developing TL & SSL hybrid models

An observation from our review is that there have been few attempts to combine the TL and SSL methods to take advantage of both approaches. For instance, Azizi's study (Azizi et al., 2023) used a TL model with ImageNet weight as a feature extractor before pre-training an SSL model, allowing the SSL model to start with more valuable and meaningful features. Although the evaluation of their hybrid models has shown promising results in medical and environmental fields, it is important to note that the current number of such studies is limited and most of them are generally in the early stages of development. Notably, one concern when using complex hybrid models is the potential burden in training time and computational cost. It needs more attention and effort to achieve the right balance between model complexity and computational resources. Thus, it is a good research direction for researchers to explore the effectiveness and design more hybrid approaches in different domains, to further improve the model performance and optimise them for various domains and tasks.

Developing lightweight frameworks

A common observation from previous research is the emergence of lightweight frameworks. As more efforts are put into collecting practical data, datasets continue to grow in size, particularly in domains like NLP and environmental sciences, and the primary motivation for using pre-trained methods is not just to address data scarcity but also to expedite training speed and enhance feature extraction for improved model performance. In such scenarios, lightweight frameworks are designed to reduce the number of model parameters, thereby improving the model's efficiency in reusing training data. Additionally, lightweight models are essential for real-world applications because they reduce the computational burden, making them more suitable for deployment on mobile devices and other resource-constrained platforms. However, it is worth noting that the development of lightweight frameworks has been relatively limited and has primarily been explored in a few specific areas. This suggests that there is ample room for researchers to explore this direction in the future.

Developing ensemble frameworks

Many of the studies we reviewed have aimed to design ensemble frameworks that incorporate multiple techniques, such as domain adaptation, adversarial training, data distillation, etc. The primary goal of building these ensemble models is to address the limitations of pre-trained methods effectively. For instance, domain adaptation techniques are used to measure the similarity between domains and reduce the feature distance between the source and target domains. This technique is specifically employed to mitigate the risk of domain mismatch and negative transfer, making the pre-trained model more robust and adaptable to different target domains. Generally, ensemble frameworks have proven to be extremely valuable in improving the

efficiency of pre-trained models, enhancing their performance, and generalisation capabilities across different domains. Nevertheless, there is same issue occurs when designing an ensemble framework as it happens in hybrid model designing, for those methods add to the model's complexity, putting extra burden on the training resources and expense, which require more experiments to achieve a balance in training cost and model performance. Thus, this can be a good direction for researchers to explore in the future.

Avoiding domain mismatch

TL approaches have seen successful applications across various domains and have become standard practice in many research areas. However, a significant challenge in the TL method is the potential risk of negative transfer between domains. This challenge has posed obstacles to the advancement of TL because quantifying and evaluating the differences between the source domain and the target domain can be complex and challenging. The performance of TL approaches is heavily dependent on the quality of the features extracted by the model. Even minor differences at the pixel level between features can have a substantial impact on the final results. Therefore, it is crucial for future research to focus on finding ways to reduce disparities between domains or mitigate the influence of domain mismatch.

Improving explainable capability

Another significant gap in the applications of the TL method that requires improvement in the future is related to the reliability of TL models. Trustworthy research in the field of deep learning has gained increasing attention in recent years. Understanding how a model makes decisions and where it pays attention is crucial for evaluating its reliability before deploying it in real-world applications. However, based on our observations, there are few approaches in our review that have applied additional explainable attention techniques and attempted to analyse their models' decision-making processes. This issue holds particular importance in professional fields such as medicine and industry, where ensuring the model's explainability is essential to building a trustworthy application in the real world. Therefore, future research should aim to address this insufficiency by including more explainable AI techniques when designing TL models.

Developing domain-specific pretext tasks

There is a gap in SSL mentioned in the review, as there are no sufficiently well-designed pretext tasks for each field. This is a significant issue for SSL applications because the SSL model learns features and representations by training source data to solve pretext tasks. The design of the pretext task may significantly influence the final performance of the entire SSL model. However, most of the available pretext tasks are based on general data types such as images and text, which makes it difficult for researchers to choose suitable pretext tasks for their models. Thus, it can be a good research direction to design task-specific or domain-specific pretext tasks to improve the efficiency of utilising training data and pretraining.

Utilising multi-modal learning

A notable observation in SSL applications is the limited research focused on multi-modal data settings. As previously discussed, SSL models can effectively utilise data from various modalities and sources, which offers a significant advantage in extracting diverse features and enhancing the model's generalisation capability. However, there are a relatively small number of studies that have delved into the utilisation of multimodal data in training SSL models. The lack of related research may be caused by the complexity of combining different data types and building fusion models, leaving room for further exploration and understanding of how different aspects of training data influence the performance of the SSL model. In future work, researchers can explore the impact of the diversity, size, distribution, and characteristics of training data on SSL models. By conducting more comprehensive investigations into the use of multimodal data, researchers can gain insights into the optimal strategies for collecting and utilising training

data to enhance the performance and generalisation capabilities of SSL models across a broader range of applications and domains.

Expanding the utilisation of SSL methods

Another current gap in SSL applications is the limited range of domains explored in current research. Throughout our review, we observed that most SSL research has primarily been designed to address data scarcity issues and tackle complex downstream tasks. SSL methods possess the unique capability to extract powerful representations from unlabelled data and leverage multimodal data, which may give them a distinct advantage in dealing with different types of data. Recent SSL approaches have demonstrated their potential to achieve performance levels similar to those of supervised learning models in many fields. This suggests that there is a promising future direction for expanding the utilisation of SSL models across a broader spectrum of application areas. Researchers can explore the effectiveness of SSL methods in diverse domains and assess their potential to become a standard approach for various applications such as text processing and computer vision learning.

9. Conclusion and future work

In this review, we have provided definitions of the TL and SSL methods and compared their training differences. Furthermore, we reviewed 80 recent studies in various application areas, analysing the strategies they used and the performance they achieved. Based on our analysis of state-of-the-art studies, we highlighted the advantages and limitations of transfer learning and self-supervised learning methods. Furthermore, we provided recommendations for selecting the appropriate method in different fields of application.

Key Findings

In summarising our observations across various applications, we have identified key findings to highlight the advantages and limitations of TL and SSL methods, along with recommendations for different domains.

1. This study encompassed a comprehensive review of the TL and SSL applications over the past four years (2020–2023). We observed a wide use of TL and SSL methods in the fields of medicine, industrial, remote sensing, natural language processing, motion, environment, Web, speaker, vehicle, robot, and physics.
2. The TL method offers several advantages, particularly in image-based domains. These advantages include mitigating data scarcity, preventing overfitting, accelerating convergence speed, improving data reuse efficiency and the quality of extracted features, supporting simulation training, reducing training time, and improving generalisation capability.
3. The TL method has several main limitations including the risk of domain mismatch, negative knowledge transfer, and the risk of label shift. Some other limitations such as heavy reliance on large-scale pre-training data, the absence of effective data quality handling techniques, and the lack of utilising explainable model techniques. Acknowledging and addressing these limitations is crucial to optimise the application of TL in various domains and scenarios.
4. The SSL method provides certain advantages, including its ability to learn from unlabelled data and alleviate data scarcity, improve feature learning, and improve model generalisation capability. Moreover, SSL provides unique benefits such as saving manual labelling efforts, utilising multi-modal data, and supporting unsupervised continuous learning.
5. The SSL method shared some limitations similar to the TL method, such as the risk of domain mismatch, the reliance on large-scale pre-training data, and the lack of combining explainable techniques. Furthermore, SSL methods have their own limitations, such as requiring specific designed pretext tasks for different domains, lack of strong supervision signals, higher computing cost than the TL method, and risk of low reliability in pseudo-label.

6. We provide recommendations for pre-trained methods across different domains. Specifically, we advocate for the use of TL in the industrial, environmental, and vehicle domains. In contrast, we endorse SSL for the medical, remote sensing, motion, web, speaker, and robot domains.

Limitations

There are two limitations in our work, which can be shown as follows:

1. In the application analysis section, fields like vehicle, physics, and robot contain a relatively small number of works, which may influence the reliability of the following analysis of methods' advantages and limitations and the field recommendations.
2. In this review, we have focused primarily on commonly used data types such as images, audio, text, and video. We included only one study that was conducted with tabular data and did not explore the effectiveness of other data types beyond these categories.

Future Direction

Future work for TL and SSL methods includes further expanding their unique advantages and avoiding current gaps.

1. Developing more TL and SSL hybrid models to take advantage of both methods.
2. Developing lightweight frameworks for the TL and SSL approaches to better adapt to mobile device applications and reduce the computational burden.
3. Developing ensemble frameworks for TL and SSL models to further enhance the model's performance.
4. Exploring more efficient ways to reduce the disparities between domains, to further avoid domain mismatch and negative knowledge transfer.
5. Enhancing the interpretability and explainability of TL models, enabling users to understand the reasoning behind model decisions.
6. Developing domain-specific pretext tasks for SSL methods to improve feature extraction quality.
7. Utilising multi-modal learning in SSL applications to increase the model's generalisation capability and understand the nature of different data types.
8. Expanding the utilisation of the SSL method in wider domains to explore more potential.
9. Investigation of the risk of unfair data in both the TL and SSL.

CRedit authorship contribution statement

Zehui Zhao: Conceptualization, Validation, Writing – original draft. **Laith Alzubaidi:** Conceptualization, Resources, Formal analysis, Writing – review & editing, Validation, Supervision. **Jinglan Zhang:** Writing – review & editing, Formal analysis, Supervision. **Ye Duan:** Formal analysis, Writing – review & editing. **Yuantong Gu:** Writing – review & editing, Validation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

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